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# Reconstructing paleoenvironments of the Late Cretaceous Western Interior Seaway, USA, using paired triple oxygen and carbonate clumped isotope measurements

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## ABSTRACT

**Fossiliferous carbonate concretions are commonly found in sediments deposited in the Late Cretaceous Western Interior Seaway. Although concretions are diagenetic features, well-preserved fossils from within them have been instrumental in reconstructing the temperature and $\delta^{18}O$ value of Western Interior Seaway seawater, which is essential for accurate reconstruction of Late Cretaceous climate. Here, we constrain formation conditions of Late Campanian and early Maastrichtian carbonate concretions by combining triple oxygen isotope measurements with carbonate clumped isotope paleothermometry on different carbonate phases within the concretions. We measured both fossil skeletal aragonite and sparry calcite infill from cracks and within macrofossil voids to evaluate differences between "primary" and "altered" geochemical signals. Based on the two temperature-sensitive isotope systems of the primary fossil shell aragonite, the temperature of the Western Interior Seaway was between 20 °C and 40 °C and was likely thermally stratified during the Campanian. The reconstructed $\delta^{18}O_{seawater}$ values of ~−1‰ for Campanian Western Interior Seaway waters are similar to those expected for the open ocean during greenhouse climates, while the Maastrichtian Western Interior Seaway may have been more restricted, with a $\delta^{18}O_{seawater}$ value of ~2‰, which reflects more evaporative conditions. We reconstructed the diagenetic history of the sparry infill and altered fossils using a fluid-rock mixing model. Alteration temperature, alteration fluid $\delta^{18}O$ value, and the initial formation temperature were calculated by applying the fluid-rock mixing model to a particle swarm optimization algorithm. We found a different range of initial formation temperatures between the Campanian (25–38 °C) and Maastrichtian (9–28 °C). We also found that alteration in the presence of light meteoric fluids ($\delta^{18}O \approx -10$‰) is required to explain both the sparry infill and the altered fossil isotopic values. Based on our results, both lithification and alteration of the carbonates occurred soon after burial, and light meteoric fluids support prior findings that high-topographic relief existed on the western margin of the Western Interior Seaway during the Late Cretaceous. As one of the first studies to apply these techniques in concert and across multiple mineralogical phases within samples, our results provide important constraints on paleoenvironmental conditions in an enigmatic ocean system and will improve interpretations of the overall health of ecosystems leading into the end-Cretaceous mass extinction.**

## INTRODUCTION

Atmospheric $CO_2$ levels during the Early Cretaceous were four to seven times higher than today, resulting in a generally warmer greenhouse climate (Arthur et al., 1985; Royer et al., 2001; Tierney et al., 2020) and a reduced global latitudinal temperature gradient (Hay, 2008; O'Brien et al., 2017). The climate during the Late Campanian and Maastrichtian stages of the Late Cretaceous reflects a general cooling trend (Barrera and Savin, 1999; Clarke and Jenkyns, 1999; Huber et al., 1995, 2002; Tierney et al., 2020), and associated strengthening of global latitudinal temperature gradients (O'Brien et al., 2017) prior to the mass extinction at the Cretaceous–Paleogene boundary. In North America, fossiliferous sediments deposited in the shallow epicontinental Western Interior Seaway contain a spectacular record of life and environments during this time period (Kauffman and Caldwell, 1993; Roberts and Kirschbaum, 1995). Nevertheless, accurately reconstructing oceanographic and climatic conditions remains a challenge, even though this information provides important constraints on the vulnerability of marine and terrestrial ecosystems leading into a major biotic transition (Brusatte et al., 2015; Condamine et al., 2021; Landman et al., 2015; Sakamoto et al., 2016; Stinnesbeck et al., 2012; Wang and Dodson, 2006).

The Western Interior Seaway underwent a regression in the Late Cretaceous following global sea-level trends and uplift along its western margin, with offshore depths shallowing to <100 m (Gill and Cobban, 1966), compared to 300 m during the mid-Cretaceous highstand (Miall et al., 2008). Over this period, the connectedness of the Western Interior Seaway with the open ocean to the north and south is debated (Hay et al., 1993; Kauffman, 1984; Kauffman and Caldwell, 1993; Kennedy et al., 1998; Petersen et al., 2016), which is an important consideration when trying to estimate oceanographic conditions in the seaway and regional climate reconstruction in general. Kauffman (1984) and Hay et al. (1993) proposed that during the Campanian, the Western Interior Seaway was connected to the Gulf of Mexico, Arctic Ocean, and Hudson Bay. This open seaway would have been a well-mixed, "normal marine" water body with an oxygen isotope composition similar to the overall ocean, perhaps similar to the present-

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day South China Sea (Schmidt et al., 1999). Coulson et al. (2011), Kennedy et al. (1998), and others have proposed a much more restricted seaway during the Campanian and subsequent Maastrichtian stage, with little connection to the north. A more restricted seaway could have been isotopically distinct from the open ocean, for instance, featuring a higher oxygen isotope value of seawater ($\delta^{18}O_{seawater}$) due to increased evaporation, similar to the present-day Mediterranean Sea (Schmidt et al., 1999). Alternatively, Petersen et al. (2016) and others have proposed variable $\delta^{18}O_{seawater}$ values for the Western Interior Seaway due to density stratification driven by large salinity gradients, similar to the present-day Chesapeake Bay (Sachs and Schwab, 2011). Density stratification could have been caused by a northern influx of freshwater into the evaporation-dominated, saline, and restricted Western Interior Seaway from two possible precipitation sources: (1) high-elevation runoff from the western mountain range, where the $\delta^{18}O$ value for precipitation is estimated to be as low as −20‰ (Fricke et al., 2010), and (2) mid- to low-elevation basinal precipitation (with $\delta^{18}O$ values of about −9.5‰; Poulsen et al., 2007).

Previous studies of Western Interior Seaway skeletal biogenic carbonates have mostly used $\delta^{18}O$ measurements to reconstruct paleotemperatures (Cochran et al., 2003; He et al., 2005; Landman et al., 2018a; Rowe et al., 2020; Tourtelot and Rye, 1969; Witts et al., 2020; Wright, 1987). However, the calculation to reconstruct temperature requires an assumed $\delta^{18}O$ value for the water ($\delta^{18}O_{seawater}$) value in which the carbonate formed. For most of these studies, an assumption has been that Western Interior Seaway waters were isotopically similar to the open ocean, which under ice-free Cretaceous "greenhouse" conditions, would have an average $\delta^{18}O_{seawater}$ value of −1‰ (Shackleton and Kennett, 1975). However, this $\delta^{18}O_{seawater}$ value may not be an appropriate assumption for an epeiric seaway such as the Late Cretaceous Western Interior Seaway due to the oceanographic and tectonic uncertainties detailed above. Several recent studies used carbonate clumped isotopes (denoted by the isotope ratio $\Delta_{47}$) to reconstruct paleotemperature and $\delta^{18}O_{seawater}$ values of the Western Interior Seaway through the Late Cretaceous (Dennis et al., 2013; Gao et al., 2021; Jones et al., 2022; Petersen et al., 2016; Tobin et al., 2014). Carbonate $\Delta_{47}$ values are a measure of the $^{13}C$ and $^{18}O$ "clumping" in carbonate C-O bonds, which is dependent on temperature and yields paleotemperatures that are independent of mineral and precipitating fluid $\delta^{18}O$ values. This allows for an independent estimation of both temperature and the $\delta^{18}O$ value of the water from carbonate $\Delta_{47}$ and $\delta^{18}O$ (Eiler, 2011; Huntington and Petersen, 2023). However, like carbonate $\delta^{18}O$, $\Delta_{47}$ is susceptible to resetting during diagenesis and is also affected by so-called "solid state bond reordering" at elevated temperatures, which a fossil shell may experience during deep–moderate sedimentary burial (Dennis and Schrag, 2010; Hemingway and Henkes, 2021; Henkes et al., 2014; Looser et al., 2023; Passey and Henkes, 2012; Ryb and Eiler, 2018; Shenton et al., 2015; Stolper et al., 2018). Although screening tests for diagenetic alteration are used to identify the most pristine fossil samples in these studies (e.g., Cochran et al., 2010), a recent paper using carbonate triple oxygen isotopes (paired $\delta^{17}O$-$\delta^{18}O$ analyses, reported as $\Delta'^{17}O$ values) of fossil shells demonstrated diagenetic alteration of samples previously assumed to be pristine based on luminescence, microstructure, and elemental (Sr, Fe, and Mn) compositions (Wostbrock et al., 2020a). The triple oxygen isotope values are particularly sensitive indicators of diagenesis because both $\delta^{17}O$ and $\delta^{18}O$ values have identical diagenetic susceptibility (in contrast to a different potential for carbonate $\delta^{13}C$ because of varying dissolved inorganic carbon concentrations in fluids; Banner and Hanson, 1990). There is a unique paired equilibrium $\delta^{17}O$-$\delta^{18}O$ value of a carbonate for a given temperature in ocean water with a given $\delta^{18}O$ value, which means that deviations from the expected equilibrium value are easily identifiable.

In this study, we paired $\Delta'^{17}O$ and $\Delta_{47}$ measurements along with traditional $\delta^{13}C$ and $\delta^{18}O$ measurements of carbonates found within nine concretions spanning the Late Campanian and early Maastrichtian in the Western Interior Seaway. Multiple phases and mineralogies of carbonate occur in these concretions. Phases include crack-filling sparry calcite, void-filling sparry calcite, aragonitic fossil shells of ammonoid cephalopods, and both aragonitic and calcitic fossil shells of bivalve mollusks. By targeting multiple phases within a single concretion, we further demonstrate that it is possible to "see through" diagenesis and constrain the initial precipitation temperature of the biogenic carbonates (Wostbrock et al., 2020a; Wostbrock and Sharp, 2021). By measuring both $\Delta'^{17}O$ and $\Delta_{47}$, we can better control for the effects of diagenesis and calculate paleotemperatures and $\delta^{18}O_{seawater}$ of the Western Interior Seaway using both pristine samples and altered samples. The additional application of fluid-rock mixing models helps to constrain the initial conditions of carbonate formation in altered samples. These results are significant because they both enable the identification of diagenetic effects on common geochemical proxies used to characterize paleoenvironments and provide insight into conditions in one of the most well-studied geologic basins leading into one of the largest environmental perturbations of the Phanerozoic.

## METHODS

### Sample Selection and Formation Information

Samples of fossiliferous concretions were selected for isotopic analysis to represent a variety of contemporaneous or near-contemporaneous oceanographic settings, such as nearshore versus offshore and near methane seep versus non-methane seep (Table 1). Concretions from the Western Interior Seaway are often fossil-rich and are commonly found in continuous layers within friable shales (Reeside and Cobban, 1960; Waage, 1964). In some cases, these layers may represent subtle flooding surfaces (i.e., parasequence boundaries; Slattery et al., 2018). They are believed to have formed during early diagenesis after rapid burial (Landman and Klofak, 2012; Maples, 1986; Zatoń and Marynowski, 2006). Maples (1986) addressed four main identifying features that indicate an early diagenetic origin: (1) The surrounding shale tends to drape over concretions; (2) fossils are not crushed, so they must have been lithified into the concretion before further burial; (3) fossils tend to be better preserved in the concretions compared to the shale; and (4) bioturbation can be preserved in the concretions but not observed in the host shale. Some studies suggest that concretions tend to form where clusters of mollusks adhered or were lying on the seafloor (Waage, 1964, 1968). Alternatively, shells may have accumulated preferentially in depressions on the seafloor that were either created biologically (e.g., depressions created from feeding rays; cf. Geraghty and Westermann, 1994) or from bottom water currents (Tsujita, 1995; Zatoń and Marynowski, 2006), or a mixture of the two (Landman and Klofak, 2012). However, it is worth noting that not all concretions are fossil bearing, which suggests that the presence of organic matter, not necessarily from fossils, may be necessary for concretions to form (Raiswell and Fisher, 2000, 2004).

Landman and Klofak (2012) proposed a timeline of carbonate concretion formation that occurs in three steps. First, over a period of less than 10 years, ammonite and other mollusk shells accumulate in localized areas due to predation and/or currents. Over the next 10–100 years, the concretion forms after shallow burial with concomitant calcite formation within cracks. Finally, alteration by meteoric water during later diagenesis results in the altered fossil shell material and infilling of ammonite



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TABLE 1. SAMPLE INFORMATION FOR CONCRETIONS USED IN THIS STUDY

| Sample ID | Type | Age (Ma) | Taxon | Biozone | Stratigraphy | Locale | Latitude/Longitude (°N/°W) | Locale number | Habitat |
|---|---|---|---|---|---|---|---|---|---|
| UNM15502-1 | Shell | 73.8 | *Hoploscaphites* sp. | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15502-2 | Infill | 73.8 | N.A. | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15502-4 | Shell | 73.8 | Inoceramid bivalve | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15503-1 | Shell | 73.8 | *Placenticeras meeki* | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15504-1 | Shell | 73.8 | *Nymphalucina occidentalis* | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15504-2 | Shell | 73.8 | Inoceramid bivalve | *B. compressus* | Pierre Shale Fm | Patterson Ranch, South Dakota | 44.17306/102.40194 | AMNH 3528 | O/S |
| UNM15473-1 | Shell | 73.8 | *Baculites compressus* | *B. compressus* | Pierre Shale Fm | Trask Ranch, South Dakota | 44.1359/102.2412 | AMNH 3966 | O/NS |
| UNM15505-1 | Shell | 73.8 | *Baculites compressus* | *B. compressus* | Pierre Shale Fm | Smith Ranch, South Dakota | 44.2375/102.5761 | UNM 1706A | O/NS |
| UNM15505-2 | Infill | 73.8 | N.A. | *B. compressus* | Pierre Shale Fm | Smith Ranch, South Dakota | 44.2375/102.5761 | UNM 1706A | O/NS |
| UNM15505-3 | Infill | 73.8 | N.A. | *B. compressus* | Pierre Shale Fm | Smith Ranch, South Dakota | 44.2375/102.5761 | UNM 1706A | O/NS |
| UNM15526-1 | Shell | 69 | *Hoploscaphites melloi* | *H. birklundae* | Pierre Shale Fm | White Ranch, South Dakota | 44.401487/102.102057 | AMNH 3294 | O/NS |
| UNM15525-3 | Shell | 69 | Unknown bivalve | *H. birklundae* | Fox Hills Fm | Red Bird, Wyoming | 43.31998/104.25912 | AMNH 3589 | N/NS |
| AMNH46620-1 | Shell | 68 | *Hoploscaphites nicolletii* | *H. nicolletii* | Pierre Shale Fm (Elk Butte Mbr) | Moreau River, South Dakota | 45.213317/101.512167 | AMNH 3302 | O/NS |
| AMNH161248-4 | Shell | 67 | *Hoploscaphites* sp. | *H. nebrascensis* | Fox Hills Fm (Timber Lake Mbr) | Corson and Dewey Counties, South Dakota | 45.43483/101.35 | AMNH 3272 | N/NS |
| AMNH161248-1 | Infill | 67 | N.A. | *H. nebrascensis* | Fox Hills Fm (Timber Lake Mbr) | Corson and Dewey Counties, South Dakota | 45.43483/101.35 | AMNH 3272 | N/NS |

*Note:* N.A.—not applicable; Fm—Formation; Mbr—Member; O—offshore; S—seep; NS—non-seep; N—nearshore.

chambers and voids with calcite. The dissolution and reprecipitation of different carbonate phases in these concretions make them ideal samples for reconstructing the fluid that altered the materials within the concretion and calculating the pristine carbonate oxygen isotope composition. Also, the presence of discrete diagenetic phases provides an ideal scenario for using both clumped and triple oxygen isotope geochemistry to elucidate primary versus diagenetic preservation.

For this study, concretions were collected from the Pierre Shale and Fox Hills formations, which were deposited in the epeiric Western Interior Seaway during the Campanian and the early–mid Maastrichtian, respectively, and which today crop out in Kansas, Colorado, Wyoming, South Dakota, North Dakota, and Montana, USA (Fig. 1; Kauffman and Caldwell, 1993; Waage, 1968). These formations represent a range of offshore through deep subtidal to littoral depositional environments, although even in the most offshore settings, water depths in the Western Interior Seaway were <100 m (Gill and Cobban, 1966). The fossiliferous nature of the Pierre Shale and Fox Hills Formation has allowed for development of a high-resolution biostratigraphic and chronostratigraphic framework for the Late Cretaceous Western Interior Seaway based on molluscan macrofossils (primarily ammonoid cephalopod and inoceramid bivalve mollusks) and radiometric dating of volcanic ash horizons (bentonites; Cobban et al., 2006).

Of the nine concretions from this study, five Late Campanian concretions come from the *Baculites compressus* zone of the Pierre Shale in South Dakota. This biozone was dated to ca. 73.8 Ma by Landman et al. (2018b). Abundant methane cold seeps occurred in the Western Interior Seaway during this time interval (Gao et al., 2021; Landman et al., 2018a; Rowe et al., 2020), and samples were selected from both a known methane seep and non-seep locality for comparison. All *B. compressus* zone samples are from offshore localities 150–250 km east of the Western Interior Seaway shoreline based on the reconstructions of Cobban et al. (1994).

The remaining four Maastrichtian concretions come from the *Hoploscaphites birklundae, Hoploscaphites nicolletii*, and *Hoploscaphites nebrascensis* zones of both the Pierre Shale and the Fox Hills Formation in South Dakota and Wyoming. These samples were selected to represent both nearshore (Fox Hills Formation) and offshore (Pierre Shale) deposition during these time intervals, which can be dated to ca. 69–67 Ma (Hicks et al., 1999; Slattery et al., 2015). The shoreline of the Western Interior Seaway migrated toward the basin during the Maastrichtian as indicated by the time-transgressive facies of the Fox Hills Formation (Landman et al., 2013; Landman and Waage, 1993), while offshore deposition (the Pierre Shale) continued in the center of the basin (Kennedy et al., 1998). See Table 1 for detailed information about the stratigraphy and locations of each sample.

In preparation for geochemical analysis, fossil shell, sparry calcite infilling cracks and voids, and matrix-derived carbonate were identified by eye or using a bifocal microscope (Fig. 2; Fig. S1 in the Supplemental Material[1]). Fossil shell and void-filling and crack-filling sparry calcite were removed from the concretion by manual scraping using a scalpel. They were then reacted with 10% hydrogen peroxide at room temperature for 2–15 min until vigorous bubbling stopped for removal of potential organic matter. The sample was subsequently rinsed three times in distilled water and then exposed to 5% sodium hypochlorite solution for less than 4 h for additional removal of recalcitrant organic matter. The sample was again rinsed three times in distilled water and dried at 60 °C.

Mineralogical data were obtained through powdered X-ray diffraction (XRD) performed using a Rigaku D/teX X-ray diffractometer at the University of New Mexico. Jade® whole pattern fitting software was used to distinguish calcite and aragonite.

[1]Supplemental Material. Supplemental figures and discussion. Please visit https://doi.org/10.1130/GSAB.S.26006965 to access the supplemental material; contact editing@geosociety.org with any questions.



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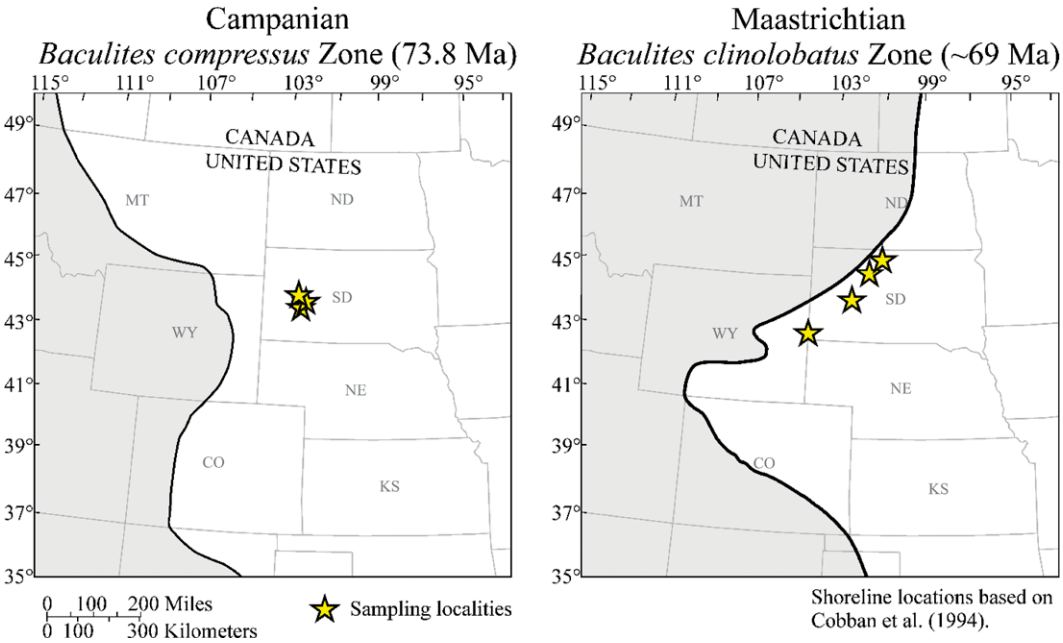


**Figure 1. Maps show sample localities in the USA. Solid black lines represent reconstructed shorelines from Cobban et al. (1994) for the Campanian *Baculites compressus* and Maastrichtian *Baculites clinolobatus* zones. White areas represent marine conditions. Gray shaded areas represent land. (A) Campanian sample locations (yellow stars) from the offshore Pierre Shale formation in South Dakota. (B) Maastrichtian sample locations (yellow stars) from the nearshore Fox Hills formation and the offshore Pierre Shale formation.**

## Stable Isotope Analysis

The $\delta^{13}C$, $\delta^{17}O$, and $\delta^{18}O$ values are reported in per mil (‰) using standard δ-notation (McKinney et al., 1950):

$$\delta^{13}C \text{ or } \delta^{x}O = \left(\frac{R_{sample}}{R_{standard}} - 1\right) \times 1000, \qquad (1)$$

where $x$ is either mass 17 or 18, and $R$ is $^{13}C/^{12}C$ or $^{x}O/^{16}O$. Oxygen isotope values are reported on the Vienna standard mean ocean water (VSMOW) scale, while the $\delta^{13}C$ values are on the Vienna Pee Dee belemnite (VPDB) scale. The $\delta^{13}C$ and $\delta^{18}O$ analyses were performed both in the Center for Stable Isotopes at the University of New Mexico (CSI-UNM), Albuquerque, New Mexico, USA, and in the Department of Geosciences Stable Isotope Laboratory at Stony Brook University (SBU), Stony Brook, New York, USA. The $\delta^{13}C$ and $\delta^{18}O$ values were obtained via conventional isotope ratio mass spectrometer (IRMS) analysis using phosphoric acid digestion (Spötl and Vennemann, 2003). For comparison of the $\delta^{13}C$ and $\delta^{18}O$ values, see Figure S10.

Triple oxygen isotope values were measured at CSI-UNM following the method described in Wostbrock et al. (2020b). In summary, five milligrams of powdered carbonate were loaded in nickel reaction vessels and reacted with bromine pentafluoride at 750 °C for four days. The oxygen ($O_2$) was quantitatively extracted and first cryogenically purified, then further purified in a 5A molecular sieve gas chromatography (GC) column before being analyzed in a Thermo Scientific MAT 253+ IRMS. The $O_2$ was analyzed relative to an in-house oxygen reference gas calibrated to VSMOW2-SLAP2 in dual inlet mode for 30 iterations with 28 s integration times. Fluorination results in more variable, and generally lower, $\delta^{18}O$ values than measurements using an IRMS connected to a GasBench; however, any potential fractionation occurring during the reaction follows a slope of ~0.528 and does not impact the $\Delta'^{17}O$ values (Wostbrock et al., 2020a, 2020b). Therefore, we used the $\Delta'^{17}O$ value obtained by fluorination and the $\delta^{18}O$ value obtained via the GasBench IRMS (Wostbrock et al., 2020a, 2020b). To calculate the $\Delta'^{17}O$ values, we linearized the $\delta^{17}O$ and $\delta^{18}O$ values (Hulston and Thode, 1965; Miller, 2002), which are denoted using $\delta'^{17}O$ and $\delta'^{18}O$ and defined as:

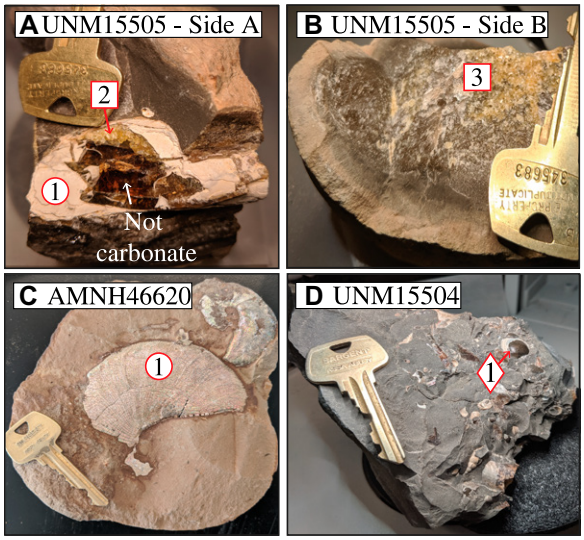


**Figure 2. Photographs of three concretion hand samples from this study (door key is shown for scale). (A, B) Concretion UNM15505, which contained an ammonite fossil (*Baculites compressus*) and two different types of sparry calcite infill. Two types of infill were sampled: void-filling calcite, which we define as calcite that precipitated inside fossil shells, and crack-filling calcite, which precipitated generally in cross-cutting cracks across the concretion. Note that the cracks in these samples do not go to the rim of the concretion. In this concretion, the void-filling calcite was only present near the fossil; a blocky, dark yellow/brown mineral (white arrow, and labeled "not carbonate"), suspected to be a sulfur-containing mineral such as barite, was also present in this void. (C) Concretion AMNH46620, which represents a concretion with no crack- or void-filling calcite. (D) Concretion UNM15504, which represents a fossiliferous concretion with different bivalve species. The arrow points to a *Nymphalucina occidentalis* bivalve fossil (sample UNM15504-1).**



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$$\delta'^{x}\mathrm{O} = 1000\ln\left(\frac{\delta^{x}\mathrm{O}}{1000}+1\right). \quad (2)$$

We used the $\delta'^{17}$O and $\delta'^{18}$O values to calculate the $\Delta'^{17}$O value using the equation (Sharp and Wostbrock, 2021):

$$\Delta'^{17}\mathrm{O} = (\delta'^{17}\mathrm{O} - 0.528\times\delta'^{18}\mathrm{O})\times 1000. \quad (3)$$

All $\Delta'^{17}$O values are reported in parts per million (ppm) on the VSMOW2-SLAP2 scale (Wostbrock et al., 2018). Since replicates were limited due to sample-size constraints, we assumed a $\Delta'^{17}$O error of $\pm 10$ ppm for all samples in this data, which was derived from the long-term $1\sigma$ standard deviation of the laboratory standards.

Carbonate clumped isotope analyses ($\Delta_{47}$) were performed at SBU, and carbonate clumped isotope temperatures, T($\Delta_{47}$), were calculated using the equation (Petersen et al., 2019):

$$\Delta_{47} = \frac{0.0383\times 10^{6}}{T^{2}} + 0.17, \quad (4)$$

where $T$ is temperature in Kelvin. $\Delta_{47}$ values are reported in per mil units, and error is reported to the 95% confidence level. The $\Delta_{47}$ measurements were made using the method described in Gao et al. (2021). Briefly, 5–7 mg aliquots of powdered carbonate were reacted with phosphoric acid at 90 °C for 10 min. The $CO_2$ was constantly collected in a liquid nitrogen trap after being passed through a −80 °C water trap. The $CO_2$ was then passed through a GC column for purification, concentrated on a cold finger, and analyzed in dual inlet mode on a Thermo Scientific MAT 253+ IRMS. Samples were analyzed against a reference $CO_2$ gas for 20 s of sample integration for 10 iterations seven times (for a total of 70 iterations), corrected using thermally equilibrated $CO_2$ gases (Dennis et al., 2011), and compared with carbonate standards, some of which have internationally accepted $\Delta_{47}$ values (Bernasconi et al., 2021).

## Fluid-Rock Mixing Model

We modeled trajectories of $\Delta'^{17}$O and $\delta^{18}$O during carbonate-water exchange due to fluid-rock interaction (diagenesis with a meteoric fluid) using the mass-balance model modified from Taylor (1978) to include $\delta^{17}$O. The bulk composition of the system is the mass balance between the initial and final water and rock oxygen isotope compositions following the equation:

$$\begin{aligned} &X_{\mathrm{water}}\left(\delta^{x}\mathrm{O}_{\mathrm{water,final}}\right) + \left(1-X_{\mathrm{water}}\right)\left(\delta^{x}\mathrm{O}_{\mathrm{rock,final}}\right) \\ &= \delta^{x}\mathrm{O}_{\mathrm{bulk}} = X_{\mathrm{water}}\left(\delta^{x}\mathrm{O}_{\mathrm{water,intial}}\right) \\ &+ \left(1-X_{\mathrm{water}}\right)\left(\delta^{x}\mathrm{O}_{\mathrm{rock,intial}}\right), \end{aligned} \quad (5)$$

where $X_{\mathrm{water}}$ is the fraction of water, and $x$ is either mass 17 or 18. The $\delta^{x}\mathrm{O}_{\mathrm{rock,final}}$ requires an estimate of the $\alpha_{\text{rock-water}}$ value, where

$$\alpha_{\mathrm{rock-water}} = \frac{1000+\delta^{x}\mathrm{O}_{\mathrm{final,rock}}}{1000+\delta^{x}\mathrm{O}_{\mathrm{final,water}}}. \quad (6)$$

The $\alpha_{\text{rock-water}}$ value varies with temperature following the equation (Wostbrock et al., 2020a):

$$1000\ln\alpha^{17}_{\mathrm{calcite-water}} = \left(\frac{2.84\times 10^{6}}{T^{2}} - 2.96\right)\left(\frac{-1.39}{T}+0.5305\right) \quad (7)$$

for oxygen-17, and

$$1000\ln\alpha^{18}_{\mathrm{carbonate-water}} = \frac{2.84\times 10^{6}}{T^{2}} - 2.96 \quad (8)$$

for oxygen-18. Note that we use "carbonate-water" in Equation 8 since the same equation applies to both aragonite-water and calcite-water (Wostbrock et al., 2020a). However, this equation may not apply to other carbonate mineralogy such as dolomite. Combining Equations 5 and 6 yields the calculation for $\delta^{x}\mathrm{O}_{\mathrm{rock,final}}$ (Wostbrock and Sharp, 2021):

$$\delta^{x}\mathrm{O}_{\mathrm{rock,final}} = \frac{1000X_{\mathrm{water}} + \alpha_{\mathrm{rock-water}}\times\left(\begin{array}{l} X_{\mathrm{water}}\delta^{x}\mathrm{O}_{\mathrm{rock,inital}} - X_{\mathrm{water}} \\ \quad \delta^{x}\mathrm{O}_{\mathrm{water,inital}} - \delta^{x}\mathrm{O}_{\mathrm{rock,inital}} \\ \quad -1000X_{\mathrm{water}} \end{array}\right)}{\alpha_{\mathrm{rock-water}}X_{\mathrm{water}} - \alpha - X_{\mathrm{water}}}. \quad (9)$$

Two equations were calculated for $\delta^{17}\mathrm{O}_{\mathrm{rock,final}}$ and $\delta^{18}\mathrm{O}_{\mathrm{rock,final}}$ values and then were converted to prime notation for $\Delta'^{17}$O calculations.

We can use the framework for this fluid-rock mixing model to back-calculate the initial temperature in which the shell mineralization occurred in altered samples. For this, we assume a time-integrated average temperature of alteration, a $\Delta'^{17}$O and $\delta^{18}$O value of the diagenetic fluid, and the $\Delta'^{17}$O and $\delta^{18}$O value of the seawater. The initial carbonate mineralogy of the shell material was aragonite. Thus, the fractionation between aragonite and water was used to calculate the initial triple oxygen isotope composition of the mineral using Equation 8 for the $\delta^{18}$O value and the $\delta^{17}$O using the aragonite-water fractionation equation from Wostbrock et al. (2020a):

$$1000\ln\alpha^{17}_{\mathrm{aragonite-water}} = \left(\frac{2.84\times 10^{6}}{T^{2}} - 2.96\right)\left(\frac{-1.53}{T}+0.5305\right). \quad (10)$$

For clarification, Equation 7, which is for calcite-water fractionation, is used to represent alteration from aragonite mineralogy to calcite mineralogy during the fluid-rock interaction, while Equation 10 is used to represent initial aragonite mineralogy.

We used a particle swarm optimization (PSO) algorithm available in the Python (3.x) pyswarm package to solve the fluid-rock mixing model (Eqs. 7–10) and found the best values of alteration temperature, alteration fluid $\delta^{18}$O, and initial formation temperature to describe the measured $\Delta'^{17}$O values. In general, a PSO uses multiple particles (referred to as a swarm) and iterates through a field of parameters (in this study: alteration temperature, alteration fluid $\delta^{18}$O, and initial formation temperature) applied to a set of equations (in this study, the fluid-rock mixing model) to minimize the residual between the modeled data and the observed (or measured) data. We used a swarm size of 100 particles and allowed the PSO to iterate up to 500 times to find the minimal residuals. The PSO usually found the optimized parameters within 10–60 iterations. Residuals were defined as the sum of the absolute difference between the measured $\Delta'^{17}$O values of our dataset and the modeled $\Delta'^{17}$O values from the fluid-rock mixing model calculated from the parameters of the PSO. The PSO iterates through the fluid-rock mixing model using different parameters until the residual differences between iterations are below a threshold value (we used the default value of 10e-8), after which the PSO considers that no more appreciable decrease in residuals will occur and outputs the optimized parameters. A PSO may provide slightly different optimized parameters after each run since the PSO stops iterating over the parameter space when the difference between iterations falls below a certain threshold. With enough runs, the PSO will have moved toward the optimized parameter from many different angles within the parameter space, giving a range of optimized parameters that all minimize the residuals of the set of equations equally. We recorded the optimized parameters from 1000 PSO runs to capture the entire range of possible optimized values that minimize the residuals of the fluid-rock mixing model.

We set each parameter's upper and lower limits so that the PSO could only choose



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values between those limits for each iteration. We set a wider range for the parameters than expected so that the PSO would not find an optimized parameter at the end of the set parameter space. For alteration temperatures, a temperature range of 0–75 °C was used. We chose this range to accommodate the range of burial temperatures the concretion may have experienced during alteration. If the alteration occurred close to the sediment-water interface, the temperature would be similar to the bottom water temperature (minimum of ~0 °C) while later alteration after burial would follow the geothermal gradient and increased temperature. We set a maximum burial temperature of 75 °C based on Denver Basin well temperatures and burial rates from Simon et al. (2023), which is near our sampling locations and taken to be roughly representative. Alteration fluid $\delta^{18}O$ values correspond with potential meteoric water $\delta^{18}O$ values (−20‰–0‰). We chose this range based on the lowest $\delta^{18}O$ value (~−20‰) suggested by Fricke et al. (2010) if there was high topography to the west of the Western Interior Seaway during the Late Cretaceous. We chose 0‰ as the highest $\delta^{18}O$ value for the alteration fluid to represent precipitation in warm, humid conditions with low topography. Lastly, we set the initial shell precipitation temperature to a range of 0–45 °C to represent a larger range than expected temperatures based on the minimum temperature of bottom water (0 °C) and the maximum temperature results from this study (maximum temperature reported is ~40 °C). We calculated the meteoric water $\delta'^{17}O$ value using Equation 8 from Aron et al. (2021) to represent the global meteoric water line ($\delta'^{17}O = 0.5268 \times \delta'^{18}O + 0.015$) and assumed a paleo-ocean $\delta^{18}O$ and $\Delta'^{17}O$ value of −1‰ and −0.002‰ (−2 ppm), respectively. Setting a wider parameter space for each parameter results in a slower model and in model output that is unrealistic given known analyses (such as maximum burial temperature; Figs. S8 and S9 provide more discussion). The $\delta^{18}O_{seawater}$ value was not used as a parameter because changing that value from +2‰ to −3‰ does not change results as much as the other three parameters (Fig. S7). More information on the PSO deployment and convergence can be found in Figures S2–S9.

## RESULTS

Four bivalve shells, seven ammonite shells, and four void/crack-filling carbonate (referred to as sparry calcite) samples were analyzed (Table 1). Based on XRD data from the shell material, most samples have a majority of aragonite as the carbonate component (67%–100%; Table 2). The exception was sample UNM15525 (an unidentified bivalve), which was mostly calcite (83%) and has a small component of silica (3%). The silica portion could not be separated from the carbonate portion of the sample before fluorination for $\Delta'^{17}O$ analysis. Therefore, the $\Delta'^{17}O$ value of sample UNM15525 has a minor silica contribution.

The $\delta^{13}C$ values of the skeletal aragonite range between −6.57‰ and +1.92‰ (VPDB; Fig. 3; Tables 3 and 4). The skeletal aragonite from the locations known to contain methane seeps generally has $\delta^{13}C$ values of <−5.4‰. The exceptions are UNM15502-1 and UNM15504-2, which are found in concretions near seeps and have $\delta^{13}C$ values of −3.33‰ and −3.78‰, respectively. Non-seep–related fossils have $\delta^{13}C$ values that range from −3.87‰ to +1.92‰. In all cases where the comparison can be made, the $\delta^{13}C$ of the void-filling and crack-filling sparry calcite is considerably lower than that of the fossil shells. The void-filling and crack-filling $\delta^{13}C$ values range from −18.34‰ to −13.60‰, regardless of whether the concretion came from a seep location or not. The low $\delta^{13}C$ value of the sparry calcite is best explained by oxidation of organic matter or formation near a methane seep where the $\delta^{13}C$ value could still be impacted. None of the $\delta^{13}C$ values are indicative of direct formation within methane seeps where anerobic oxidation of methane results in $\delta^{13}C$ values as low as −40‰ (Gao et al., 2021; Landman et al., 2018a; Rowe et al., 2020). Late Cretaceous ammonoids frequently have $\delta^{13}C$ values that are at least 4‰ lower than those of other carbonate-secreting species found in the same location regardless of proximity to known methane seeps (Landman et al., 2018a; Landman and Klofak, 2012; Tobin and Ward, 2015). Several explanations have been proposed for the lower $\delta^{13}C$ values. They could indicate that the ammonoids were not residing full time within seep conditions, resulting in an intermediate $\delta^{13}C$ value that represents both seep and non-seep conditions (Landman et al., 2018a; Landman and Klofak, 2012). Alternatively, ammonoids' rapid growth rate may require

TABLE 2. CHEMICAL COMPOSITION, EXPRESSED AS %, OF FOSSIL CARBONATE SAMPLES BASED ON X-RAY DIFFRACTION RESULTS

| Sample ID | Calcite | Aragonite | Quartz | Amorphous/Others |
|---|---|---|---|---|
| UNM15502-1 | 6.2 | 71.8 | | 22.0 |
| UNM15503-1 | | 100 | | |
| UNM15473-1 | 12.4 | 87.6 | | |
| UNM15505-1 | 32.2 | 67.8 | | |
| UNM15526-1 | 11.1 | 88.9 | | |
| UNM15525-3 | 82.7 | 13.9 | 3.3 | |
| AMNH46620-1 | | 100 | | |
| AMNH161248-4 | 10.0 | 90.0 | | |

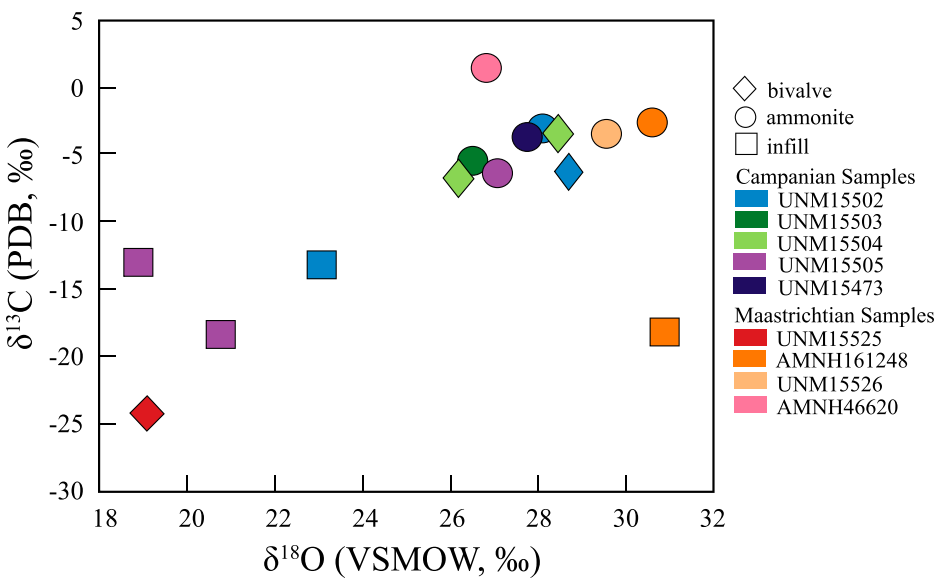


**Figure 3. Carbon and oxygen isotope data for the Western Interior Seaway samples. Fossil shells (circles—ammonoids, diamonds—bivalves) have higher $\delta^{13}C$ and $\delta^{18}O$ values than the sparry infill (squares); the exception is UNM15525, which is highly silicified (red diamond). One sparry infill sample has a higher $\delta^{18}O$ value (AMNH161248, with a $\delta^{18}O$ value of ~31‰) but $\delta^{13}C$ similar to that of the other infill samples. PDB—Pee Dee belemnite; VSMOW—Vienna standard mean ocean water.**



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TABLE 3. CARBONATE CLUMPED ($\Delta_{47}$) ISOTOPE VALUES, ALONG WITH $\delta^{13}C$ AND $\delta^{18}O$, OF CRETACEOUS FOSSIL SHELLS AND CALCITE INFILL

| Sample ID | Type | Carb type | Age (Ma) | n | $\delta^{13}C$ (‰, VPDB) | $\delta^{18}O_{carb}$ (‰, VSMOW) | $\Delta_{47}$ (‰) | $\Delta_{47}$ error (‰) | T($\Delta_{47}$) (°C) | T($\Delta_{47}$) error (°C) | $\delta^{18}O_{water}$ (‰, VSMOW) | $\delta^{18}O_{water}$ error (‰) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UNM15502-1 | Shell | Arg | 73.8 | 2 | −3.5 | 28.27 | 0.568 | 0.024 | 37 | 10 | 1.3 | 1.8 |
| UNM15502-2 | Infill | Cal | 73.8 | 3 | −13.2 | 23.66 | 0.560 | 0.017 | 42 | 7 | −2.3 | 1.3 |
| UNM15503-1 | Shell | Arg | 73.8 | 3 | −5.0 | 26.66 | 0.556 | 0.017 | 42 | 7 | 0.6 | 1.3 |
| UNM15473-1 | Shell | Arg | 73.8 | 3 | −4.1 | 27.91 | 0.574 | 0.017 | 35 | 6 | 0.5 | 1.2 |
| UNM15505-1 | Shell | Arg | 73.8 | 3 | −6.8 | 26.40 | 0.595 | 0.017 | 27 | 6 | −2.5 | 1.2 |
| UNM15505-2 | Infill | Cal | 73.8 | 3 | −13.3 | 18.15 | 0.573 | 0.021 | 35 | 8 | −9.0 | 1.3 |
| UNM15526-1 | Shell | Arg | 69 | 2 | −10.8 | 29.08 | 0.561 | 0.021 | 40 | 8 | 2.6 | 1.5 |
| UNM15525-3 | Shell | Arg | 69 | 2 | −24.5 | 18.27 | 0.561 | 0.062 | 40 | 25 | −7.9 | 4.6 |
| AMNH46620-1 | Shell | Arg | 68 | 2 | 1.8 | 27.34 | 0.585 | 0.035 | 31 | 13 | −0.9 | 2.5 |
| AMNH161248-4 | Shell | Arg | 67 | 3 | −2.4 | 30.56 | 0.573 | 0.017 | 35 | 6 | 3.1 | 1.2 |
| AMNH161248-1 | Infill | Cal | 67 | 3 | −18.2 | 31.06 | 0.608 | 0.017 | 23 | 6 | 1.0 | 1.2 |

*Note:* Measured at Stony Brook University, Stony Brook, New York, USA. T($\Delta_{47}$) was calculated using the calibration of Petersen et al. (2019). The calculated $\delta^{18}O$ value of the water was calculated by using the T($\Delta_{47}$) and measured carbonate $\delta^{18}O$ using the carbonate-water oxygen isotope equation of Wostbrock et al. (2020a), which is suitable for both aragonite-water and calcite-water. Error is reported as the 95% confidence interval. Average standard deviations for the $\delta^{13}C$ and $\delta^{18}O$ values are 0.15‰ and 0.14‰, respectively. Carb—carbonate; Arg—aragonite; Cal—calcite; n—number of replicates; VPBD—Vienna Pee Dee belemnite; VSMOW—Vienna standard mean ocean water.

TABLE 4. CARBONATE $\delta^{13}C$, $\delta^{17}O$, $\delta^{18}O$, AND $\Delta'^{17}O$ VALUES OF CRETACEOUS FOSSIL SHELLS AND CALCITE INFILL

| Sample ID | Type | Carb type | Age (Ma) | $\delta^{13}C$ (VPDB) | $\delta^{17}O_{carb}$ | $\delta^{18}O_{carb}$ | $\Delta'^{17}O$ (ppm) | T($\Delta'^{17}O$) (±10 °C) | $\delta^{18}O_{water}$ (±2‰) |
|---|---|---|---|---|---|---|---|---|---|
| UNM15502-1 | Shell | Arg | 73.8 | −3.3 | 14.646 | 28.083 | −82 | 23 | −1.6 |
| UNM15502-2 | Infill | Cal | 73.8 | −13.6 | 12.020 | 23.028 | −73 | | |
| UNM15502-4 | Shell | Arg | 73.8 | −6.6 | 14.954 | 28.670 | −81 | 24 | −0.9 |
| UNM15503-1 | Shell | Arg | 73.8 | −5.4 | 13.823 | 26.475 | −72 | 33 | −1.1 |
| UNM15504-1 | Shell | Arg | 73.8 | −6.4 | 13.646 | 26.133 | −66 | 41 | −0.3 |
| UNM15504-2 | Shell | Arg | 73.8 | −3.8 | 14.757 | 28.303 | −86 | 20 | −2.2 |
| UNM15473-1 | Shell | Arg | 73.8 | −3.9 | 14.549 | 27.917 | −95 | | |
| UNM15505-1 | Shell | Arg | 73.8 | −6.1 | 14.149 | 27.130 | −84 | 22 | −3.0 |
| UNM15505-2 | Infill | Cal | 73.8 | −13.2 | 9.905 | 18.965 | −65 | | |
| UNM15505-3 | Infill | Cal | 73.8 | −18.2 | 10.840 | 20.775 | −79 | | |
| UNM15526-1 | Shell | Arg | 69 | −3.7 | 15.483 | 29.672 | −73 | 32 | 1.7 |
| UNM15525-3 | Shell | Arg | 69 | −24.4 | 9.916 | 18.982 | −60 | | |
| AMNH46620-1 | Shell | Arg | 68 | 1.9 | 13.982 | 26.822 | −89 | | |
| AMNH161248-4 | Shell | Arg | 67 | −2.4 | 15.921 | 30.520 | −73 | 32 | 2.8 |
| AMNH161248-1 | Infill | Cal | 67 | −18.3 | 15.961 | 30.892 | −102 | | |

*Note:* Measured at the University of New Mexico. All isotopic values are reported in ‰ and relative to Vienna standard mean ocean water (VSMOW), unless indicated. Based on X-ray diffraction data, all shells are assumed to be mostly aragonite (Arg) and infill calcite (Cal). T($\Delta'^{17}O$) and $\delta^{18}O_{water}$ values were calculated for samples that plot within the equilibrium field using equations from Wostbrock et al. (2020a). Samples without temperature or water $\delta^{18}O$ calculations have been altered and no longer contain primary precipitation values, although results of a fluid-rock mixing model suggest an initial precipitation temperature of ~25 °C. $\delta^{17}O$ and $\delta^{18}O$ values are reported to the third decimal place for calculation of the $\Delta'^{17}O$ values ($\delta^{18}O$ and $\delta^{17}O$ covary, leading to the high $\Delta'^{17}O$ precision). Average standard deviations for the $\delta^{13}C$, $\Delta'^{17}O$, and $\delta^{18}O$ values are 0.23‰, 10 ppm, and 0.18‰, respectively. Carb—carbonate; VPDB—Vienna Pee Dee belemnite.

higher metabolic rates and more $CO_2$ released during respiration than for the smaller cohabitating organism (e.g., mollusks), resulting in lower $\delta^{13}C$ values of their shells (McConnaughey et al., 1997; Tobin and Ward, 2015).

The $\delta^{18}O$ values of the shells ranged from 26.48‰ to 30.86‰ (VSMOW; −4.30‰ to −0.05‰ versus VPDB, where $\delta^{18}O_{VSMOW} = 1.03092 \times \delta^{18}O_{PDB} + 30.92$ (Kim et al., 2015); Tables 3 and 4), similar to the reported range in Maastrichtian and Campanian ammonites from the Western Interior Seaway and Gulf Coastal Plain (Cochran et al., 2010; Knoll et al., 2016; Witts et al., 2020). The $\delta^{18}O$ values of the infill are generally 5‰–10‰ lower than the corresponding $\delta^{18}O$ values of the shell comparable to the sparry calcite veins and nodules from Western Interior Seaway methane seep concretions (Gao et al., 2021). The $\Delta'^{17}O$ value of samples ranged from −102 ppm to −60 ppm (Table 4; Fig. 4). Although there are no $\Delta'^{17}O$ values from the Western Interior Seaway to compare to the values in this study, modern calcitic brachiopod $\Delta'^{17}O$ values range from −96 ppm at 0 °C to −55 ppm at 40 °C, while values for aragonitic corals and sediment range from −114 ppm at 0 °C and −67 ppm at 40 °C (Wostbrock et al., 2020a). Except for sample AMNH161248, all infill components have higher $\Delta'^{17}O$ values than the corresponding shell material.

Measured $\Delta_{47}$ values range between 0.556‰ and 0.595‰ for the fossil shell samples and between 0.556‰ and 0.608‰ for the sparry infill. These data correspond to temperatures between 23 °C and 42 °C, with no obvious correlation with age or carbonate phase (Table 3; Fig. 5). The $\Delta_{47}$ values of the infill carbonate always resulted in lower calculated water $\delta^{18}O$ relative to the shell, which shows that the infill has interacted to a larger extent with a low $\delta^{18}O$ diagenetic fluid. Calculated $\delta^{18}O$ values of the formation water (Equation 8) using $\Delta_{47}$ values and assuming equilibrium isotope fractionation range from −9.0‰ to +3.1‰ (Table 3). The relationship between $\Delta_{47}$ and $\Delta'^{17}O$ values is shown in Figure 6.

We used the $\Delta'^{17}O$ and $\delta^{18}O$ values of the dataset in a PSO algorithm to determine formation temperatures and whether there are any temperature differences between the Campanian and Maastrichtian samples (Figs. 7 and 8). The calculated seawater temperature was 22–25 °C when considering the entire dataset in the PSO algorithm. Within each time period, we separated the dataset into higher and lower temperature bins based on the shell $\Delta'^{17}O$ values to capture the entire range of possible seawater temperatures. We consider $\Delta'^{17}O$ values below −70 ppm as "low temperature," which corresponds to seawater temperatures of 30 °C and below, for a $\delta^{18}O_{seawater}$ value of −1‰. This allowed us to separate samples whose $\Delta'^{17}O$ values alone suggest temperatures that are at or near the maxi-



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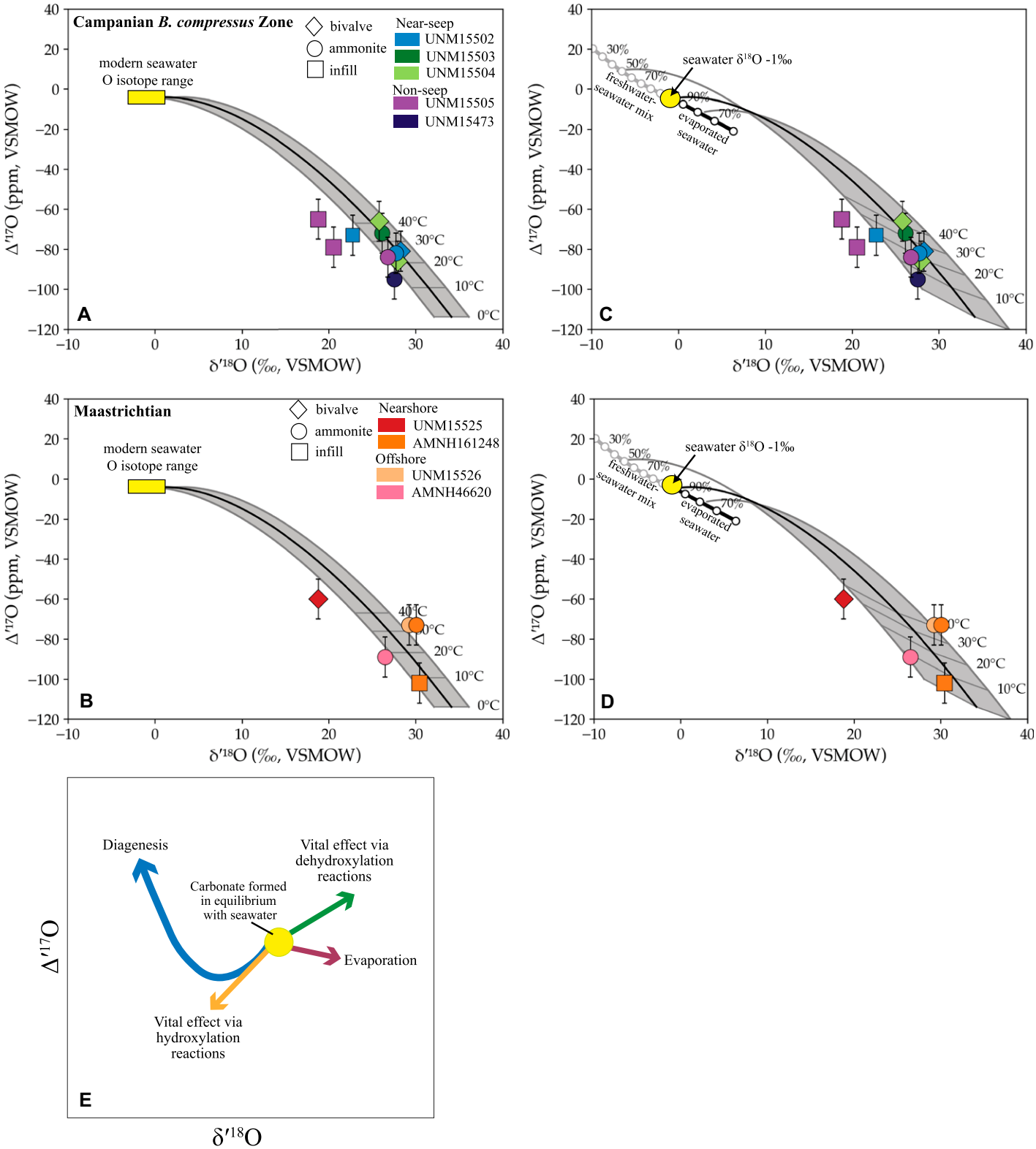


mum temperature for ectotherms (Pörtner, 2002). For the Campanian sample set, seawater temperatures range between 25 °C and 38 °C, while for the Maastrichtian sample set, seawater temperatures range between 9 °C and 28 °C.

## DISCUSSION

Here, we investigate the utility of the combined measurements of $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ along with quantitative modeling to document/calculate diagenetic alteration and increase the accuracy of geochemical reconstructions of paleoenvironmental conditions. Some carbonate concretions in this study have void-filling and crack-filling calcite, which are taken to represent



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**Figure 4. Triple oxygen isotope compositions of Late Campanian carbonate concretions for seep and non-seep localities within (A, C) the *Baculites compressus* biozone and (B, D) early Maastrichtian carbonate concretions for nearshore and offshore localities. Panels A and B use our defined Western Interior Seaway $\delta^{18}O$ values of −3‰ to +1‰, while panels C and D reflect the Western Interior Seaway $\delta^{18}O$ values required if interpreting all fossil carbonate from this study as unaltered. The aragonite-water equilibrium fractionation field is represented by the gray field with the average global ocean $\delta^{18}O$ value of −1‰ represented by the black line. (A, B) For a sample to be considered unaltered, the triple oxygen isotope composition of the fossil carbonate (circle and diamonds) must plot within the gray field that defines the triple oxygen isotope equilibrium fractionation field for aragonite-water in equilibrium with an ocean with a $\delta^{18}O$ value of −3‰ to +1‰ (yellow bar). The late-stage void-filling and crack-filling calcite samples (squares) do not plot within the equilibrium field, which suggests that alteration affected the triple oxygen isotope values. Highly silicified bivalve sample UNM15525 (red diamond) does not plot near the fractionation field, probably due to contamination. (C, D) If all fossil carbonate from this study preserved primary $\Delta'^{17}O$ and $\delta^{18}O$ values, the calculated Western Interior Seaway seawater $\delta^{18}O$ value would range from −7‰ (left gray equilibrium fractionation line) to +3‰ (right gray equilibrium fractionation line). Seawater with a $\delta^{18}O$ value of −7‰ requires the addition of 50% freshwater with a $\delta^{18}O$ value of −12‰ (gray line with white dots). Using a simple Rayleigh fractionation model to represent evaporation, 20% of the seawater would need to evaporate to produce a $\delta^{18}O$ seawater value of 3‰ (black line with white dots). (E) Schematic showing how the triple oxygen isotope composition of carbonate can be different from an expected equilibrium value (yellow circle). Evaporation results in a water with lower $\Delta'^{17}O$ and higher $\delta^{18}O$ values; therefore, a carbonate precipitating in evaporated water would have lower $\Delta'^{17}O$ and higher $\delta^{18}O$ values than expected for carbonate precipitating in equilibrium with a more normal seawater oxygen isotope composition (purple arrow). Diagenesis by infiltration of a meteoric fluid will cause the carbonate triple oxygen isotope value to change following the blue arrow. The fluid-rock ratio controls where along the blue arrow the altered triple oxygen isotope value would plot. Note that this is a general schematic, and the magnitude of change will be determined by the fluid-rock ratio, the $\delta^{18}O$ value of the meteoric water, and the temperature of alteration. Vital effects can be identified by the deviation from expected equilibrium value (green and orange arrow). Dehydroxylation of bicarbonate during carbonate formation would cause the triple oxygen isotope value to plot somewhere along the green arrow. Hydroxylation of bicarbonate during carbonate formation would cause the triple oxygen isotope value of the precipitating carbonate to plot somewhere along the orange arrow. Note that the magnitude and azimuth would change slightly depending on precipitating conditions such as pH, temperature, time, etc. See Guo and Zhou (2019) for more information on the (de)hydroxylation effects. VSMOW—Vienna standard mean ocean water.**

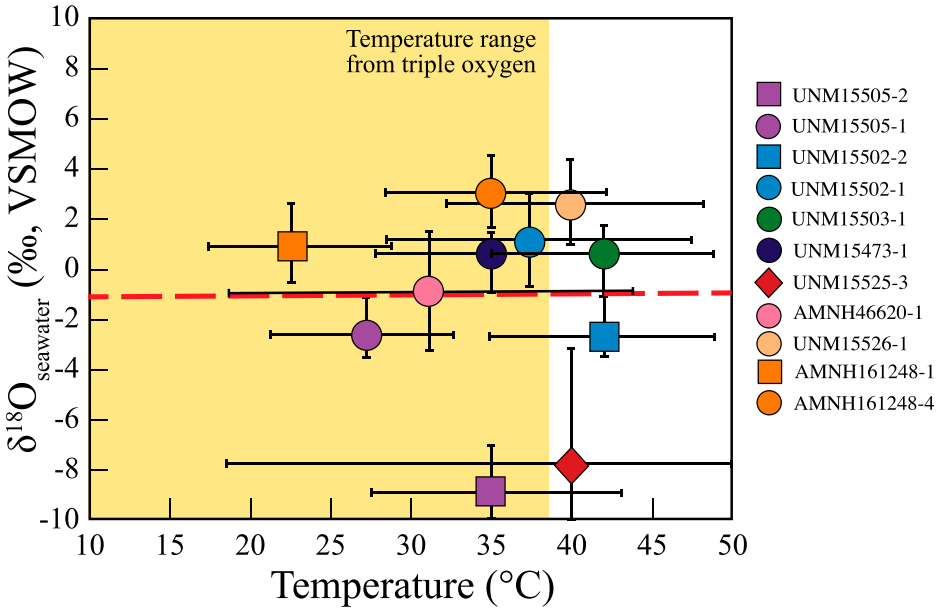


**Figure 5. Calculated $\delta^{18}O$ values of formation waters and corresponding temperatures from carbonate clumped measurements. Temperatures and calculated $\delta^{18}O_{seawater}$ values for the sparry infill (squares) overlap with those of the fossil carbonate (circles and diamond). Red dashed line represents a $\delta^{18}O_{seawater}$ value of −1‰. Most fossil carbonate samples have $\Delta_{47}$ values that correspond to calculated $\delta^{18}O_{seawater}$ values that are higher than the canonical −1‰ that represents the global average greenhouse open ocean. VSMOW—Vienna standard mean ocean water.**

a later-stage carbonate that formed from fluids (most likely meteoric water per the results of the fluid-rock mixing model) interacting with the concretion after burial. This is further inferred using the XRD data that show some original aragonite had been replaced with calcite in the shell carbonate of some originally aragonitic fossils (Table 2). We interpret the aragonite identified by XRD to most likely represent the shell that formed during ammonite growth while the presence of any calcite in the fossil represents transformation from aragonite to calcite during diagenesis. To explore the extent to which diagenesis affected the $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ of the fossil ammonite and bivalve shells, we used the $\Delta'^{17}O$ and $\delta^{18}O$ values to distinguish between pristine and altered shell samples (Fig. 4). For the altered shells, we see through the alteration by using a fluid-rock mixing model to estimate the primary precipitation temperature. Lastly, we discuss the implications of interpreting the fossil $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ values without correcting for any diagenesis.

## Using $\Delta'^{17}O$ and $\delta^{18}O$ Values to Distinguish between Pristine and Altered Samples

### *The $\delta^{18}O_{seawater}$ and $\Delta'^{17}O_{seawater}$ Values that Represent Western Interior Seaway Oceanic Conditions*

The seawater $\Delta'^{17}O$ and $\delta^{18}O$ range that we consider to be reasonable oceanic conditions during the Late Cretaceous is shown by the yellow bar in Figure 4. During a greenhouse climate with no polar ice caps present, the average $\delta^{18}O_{seawater}$ value is assumed to be −1‰ (Shackleton and Kennett, 1975), and the $\Delta'^{17}O_{seawater}$ value is assumed to be −2 ppm (McGunnigle et al., 2022), compared to an average of 0‰



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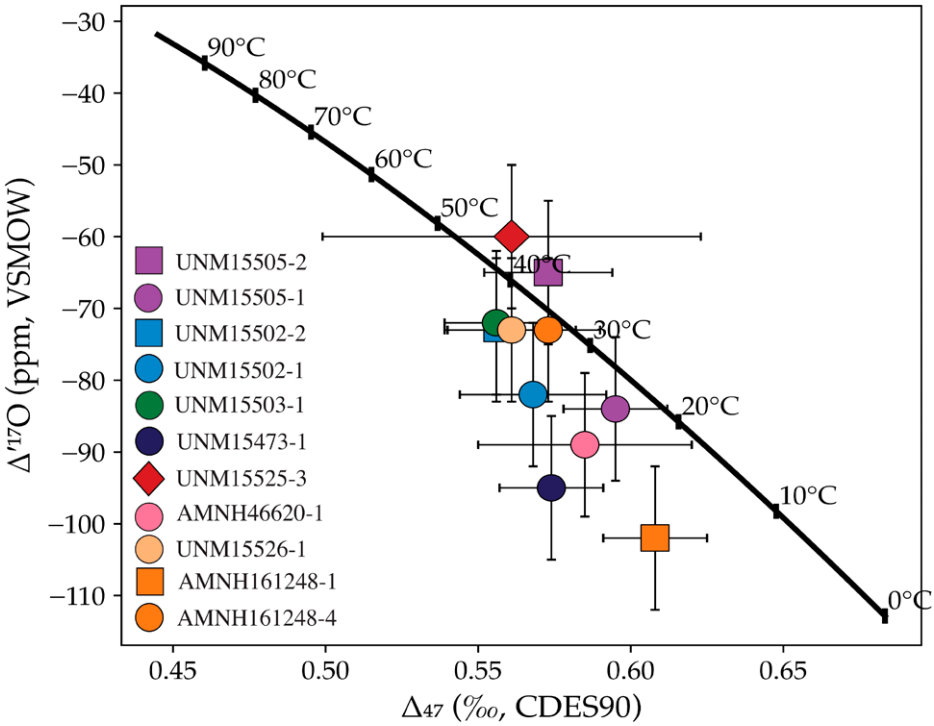


**Figure 6. The $\Delta'^{17}O$ and $\Delta_{47}$ values from Western Interior Seaway samples. Black curved line represents equilibrium for both systems with an ocean water $\Delta'^{17}O$ of −4 ppm. The $\Delta'^{17}O$-$\Delta_{47}$ equilibrium line helps visualize when a sample has preserved primary precipitation temperature or has been altered. VSMOW—Vienna standard mean ocean water.**

and −5 ppm, for $\delta^{18}O_{seawater}$ and $\Delta'^{17}O_{seawater}$, respectively, in the modern ocean where some ice caps are present (Craig and Gordon, 1965; Luz and Barkan, 2010). However, local variations are possible due to local evaporation and/or the influx of meteoric water (Kelson et al., 2022; LeGrande and Schmidt, 2006; Petersen et al., 2016; Schmidt et al., 1999). Therefore, we adapted the modern $\delta^{18}O$ range of surface seawater (−2‰ to +2‰; Schmidt et a., 1999) to greenhouse conditions (−3‰ to +1‰) to give a range that could represent Western Interior Seaway seawater (yellow line, Fig. 4). This is similar to the range of Western Interior Seaway $\delta^{18}O_{seawater}$ values calculated from $\Delta_{47}$ values in other studies (Dennis et al., 2013; Gao et al., 2021; Jones et al., 2022; Petersen et al., 2016). The corresponding $\Delta'^{17}O$ values of the endmember seawater $\delta^{18}O$ values do not change appreciably from the average seawater $\Delta'^{17}O$ value (−2 ppm). A 20% addition of meteoric water with a $\delta^{18}O$ value of −12‰ is required to change the Western Interior Seaway $\delta^{18}O_{seawater}$ value from −1‰ to −3‰. Using a canonical meteoric water $\Delta'^{17}O$ value of 0.03‰ would result in a $\Delta'^{17}O_{seawater}$ increase of only 4 ppm. On average, the $\Delta'^{17}O$ value of residual evaporated water decreases by ~2 ppm per 1‰ increase in $\delta^{18}O$ values for a change in $\delta^{18}O$ of <5‰ (Huth et al., 2022). For evaporation to cause an increase in the Western Interior Seaway $\delta^{18}O_{seawater}$ value to 1‰, the $\Delta'^{17}O_{seawater}$ value would decrease by 4 ppm to −8 ppm. The range of Western Interior Seaway $\Delta'^{17}O_{seawater}$ and $\delta^{18}O_{seawater}$ values is therefore represented by the yellow bar in Figure 4.

### *The Aragonite-Water Triple Oxygen Isotope Equilibrium Fractionation Field*

Triple oxygen isotope values of geologic samples provide information on the oxygen isotope composition and temperature of the fluid in which the sample formed because there is only one equilibrium carbonate $\Delta'^{17}O$ and $\delta^{18}O$ value for a given temperature and seawater $\Delta'^{17}O$ and $\delta^{18}O$ value. To determine whether the $\Delta'^{17}O$ and $\delta^{18}O$ values of the fossil carbonate preserve primary marine conditions or have experienced post-depositional alteration, we calculated a field

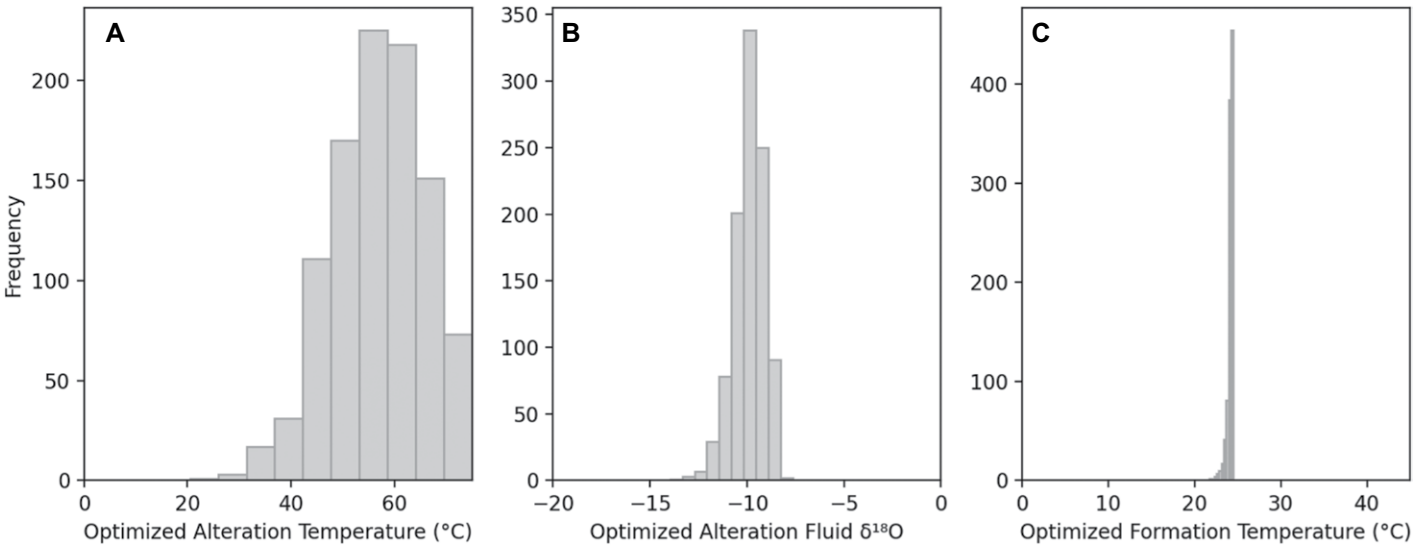


**Figure 7. Optimized parameters for (A) alteration temperature, (B) alteration fluid $\delta^{18}O$ value, and (C) formation temperature after 1000 iterations of the particle swarm optimization algorithm using all samples from the dataset.**



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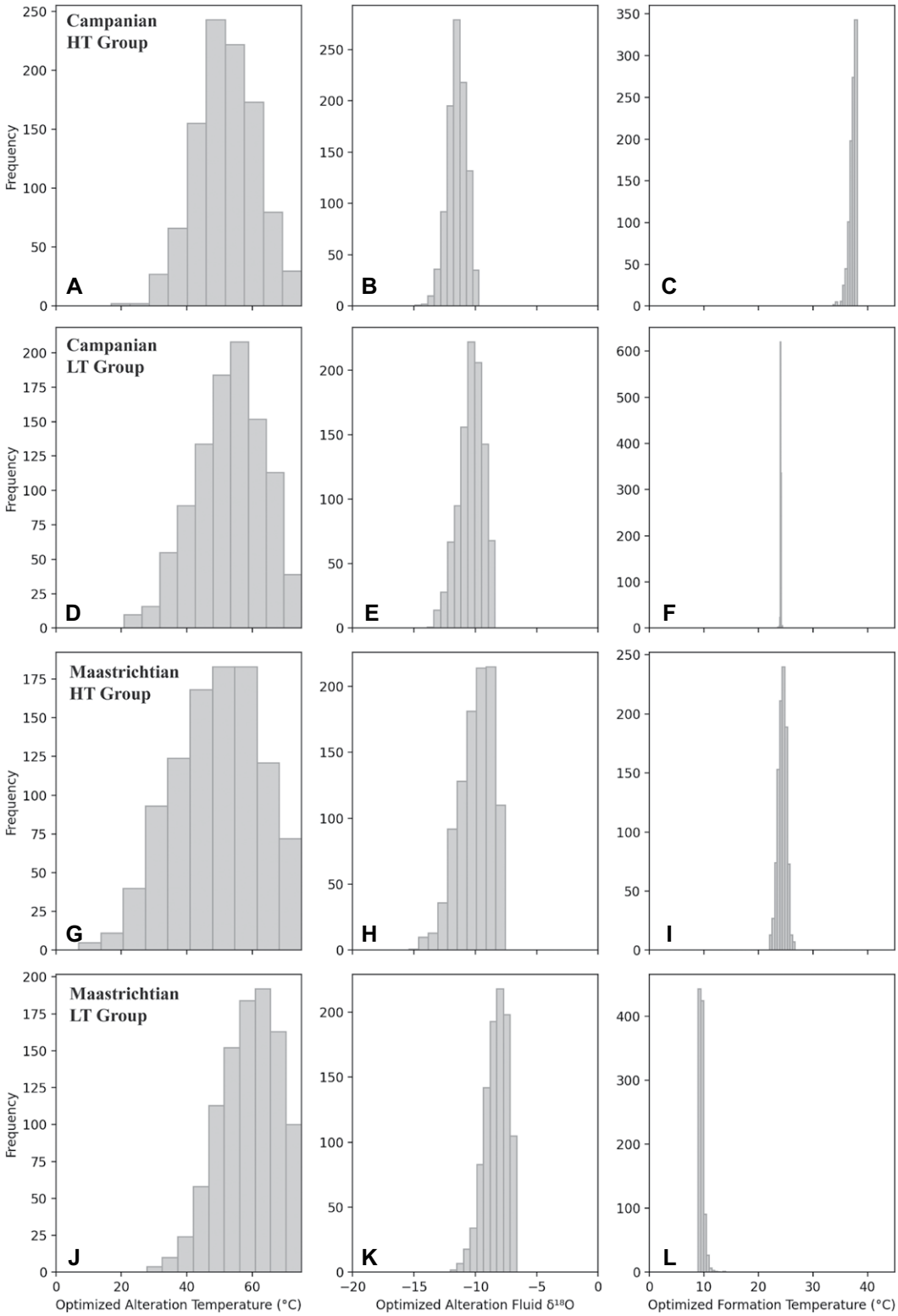


**Figure 8. Optimized parameters for (A, D, G, J) alteration temperature, (B, E, H, K) alteration fluid $\delta^{18}O$ value, and (C, F, I, L) formation temperature after 1000 iterations of the particle swarm optimization algorithm. Samples were separated into four categories: (A, B, C) high-temperature (HT) samples from the Campanian, (D, E, F) low-temperature (LT) samples from the Campanian, (G, H, I) high-temperature samples from the Maastrichtian, and (J, K, L) low-temperature samples from the Maastrichtian. See Results section in the main text for explanation of how samples were sorted into groups.**

of equilibrium carbonate triple oxygen isotope values that represent precipitation in modern seawater, adapted to a Cretaceous greenhouse climate (Fig. 4). We discuss how we calculated the seawater-carbonate equilibrium field below. Fossil carbonate $\Delta'^{17}O$ and $\delta^{18}O$ values that plot within this field are considered unaltered, or "pristine," and record the initial temperature of precipitation and the $\Delta'^{17}O$ and $\delta^{18}O$ value of the seawater in which it formed. Fossil carbonate that does not have $\Delta'^{17}O$ and $\delta^{18}O$ values that plot within this field is altered and no longer preserves the initial environmental conditions in which it formed. Its isotopic composition may, however, be used to back-calculate original conditions.



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The carbonate-water triple oxygen isotope fractionation equation was determined empirically (Wostbrock et al., 2020a) and theoretically (Guo and Zhou, 2019; Hayles et al., 2018; Schauble and Young, 2021), with the empirical estimate being within error of all of the theoretical estimates. Theoretical fractionation calculations for temperatures between 0 °C and 100 °C show an ~0.5‰ difference for the $\delta^{18}O$ fractionation and little to no difference in the $\Delta'^{17}O$ fractionation between calcite-water and aragonite-water (Schauble and Young, 2021), although the empirical results from four modern aragonite samples indicate a slightly lower $\Delta'^{17}O$ fractionation relative to calcite (e.g., 18 ppm lower at 0 °C; Wostbrock et al., 2020b). Since XRD results indicate that the biogenic carbonate is majority aragonite, we used the aragonite-water triple oxygen isotope calibration curve to describe primary precipitation environments for this dataset, which yields the equation:

$$\Delta'^{17}O_{Arg} - \Delta'^{17}O_{wt} = \left(\frac{2.84(\pm 0.02)\times 10^6}{T^2} - 2.96(\pm 0.19)\right)\left(\frac{-1.53(\pm 0.2)}{T} + 0.5305 - \lambda\right), \quad (11)$$

where $T$ is temperature in Kelvin, and $\lambda$ is the reference slope taken as 0.528 in this work (Sharp et al., 2018). Figures S12 and S13 show the dataset recalculated using the calcite-water fractionation curve from Wostbrock et al. (2020a), which results in calculated temperatures that are ~10 °C cooler than when aragonite-water is used (Equation 11).

The black line in Figure 4 shows the triple oxygen aragonite–water equilibrium fractionation line corresponding to an ocean with a $\delta^{18}O$ value of −1‰ and $\Delta'^{17}O$ value of −2 ppm. The aragonite–water equilibrium fractionation field corresponding to an ocean with $\delta^{18}O$ values of −3‰ and +1‰ is denoted by the gray field in Figure 4. If the $\Delta'^{17}O$ and $\delta^{18}O$ values of the fossil carbonate samples from this study are only a function of their habitat/formation temperature in seawater, then their $\Delta'^{17}O$ and $\delta^{18}O$ values should lie within the field in Figure 4. If a fossil carbonate sample lies outside this field, it has been altered by interaction with meteoric or pore waters (Wostbrock and Sharp, 2021). By using this range of $\delta^{18}O_{seawater}$ values, we interpret all fossil carbonate samples from this study that have $\Delta'^{17}O$ and $\delta^{18}O$ values falling between a $\delta^{18}O_{seawater}$ value of −3‰ and +1‰ as pristine and unaltered.

## Campanian Concretions (Baculites Compressus Biozone)

Based on our triple oxygen isotope results, the fossils within non-seep concretions correspond to ~1‰ lower $\delta^{18}O_{seawater}$ values than fossils within the three concretions from known methane seep localities (Table 4 and Fig. 4A). All concretions were retrieved from the Pierre Shale, which is commonly interpreted as an offshore depositional setting. Although the sample size is small, this could indicate that the higher carbon content and early cementation in sediments proximal to Western Interior Seaway cold methane seeps hinder the potential for diagenetic resetting, a phenomenon that warrants further investigation (see Landman et al., 2018a, and Gao et al., 2021).

***Offshore, non-seep concretions.*** Two shells from the ammonite *Baculites compressus* were analyzed from two different concretions (UNM15473 and UNM15505). The ammonite in concretion UNM15505 (UNM15505-1) has a triple oxygen isotope composition that corresponds to a $\delta^{18}O_{seawater}$ value of −3‰ and formed at a temperature of 22 °C (Fig. 4A). The ammonite in concretion UNM15473 has a $\Delta'^{17}O$ value that is within error of the $\Delta'^{17}O$ value of UNM15505-1 and a $\delta^{18}O$ value that corresponds to a $\delta^{18}O_{seawater}$ value of −2‰ (Fig. 4A). Concretion sample UNM15505 also contained secondary sparry infill. Void-filling sparry calcite was sampled from within a *Baculites compressus* (UNM15505-2; Fig. 2A) fossil while a crack-filling sparry calcite was sampled from within a crosscutting crack (UNM15505-3; Fig. 2B). The $\Delta'^{17}O$ and $\delta^{18}O$ values of the infill samples all indicate interaction with a light meteoric fluid (Figs. 4A and 4C). Clumped isotope temperatures for these samples range between 27 °C and 35 °C, with the sparry infill recording the highest of these temperatures (Fig. 5).

***Offshore, methane seep locality concretions.*** The three near-seep concretions contained a variety of mollusk species. UNM15503 contained a single planispiral ammonite fossil: an incomplete specimen of *Placenticeras meeki*. UNM15504 contained both lucinid and inoceramid bivalves. UNM15502 contained a scaphitid ammonite (*Hoploscaphites* sp.), inoceramid bivalves, and void-filling sparry calcite of a small indeterminate fossil. The oxygen isotope compositions of fossil shell samples consistently show equilibrium precipitation with $\delta^{18}O_{seawater}$ values between −3‰ and −1‰ (±2‰; Fig. 4A). The $\delta^{18}O$ and $\Delta'^{17}O$ values of the void-filling sparry calcite (UNM15502-2) are evidence of interaction with a light meteoric water (Fig. 4A).

Two fossils (UNM15503-1, *Placenticeras meeki*; and UNM15504-1, lucinid bivalve) have $\delta^{18}O$ and $\Delta'^{17}O$ values that correspond to a temperature of >30 °C, while the remaining fossils (UNM15502, *Hoploscaphites* sp., and two inoceramid bivalves) have $\delta^{18}O$ and $\Delta'^{17}O$ values that correspond to a temperature between 20 °C and 25 °C (Fig. 4A). The difference in temperatures could be related to habitat. *Placenticeras meeki* may have lived higher in the water column (>30 °C) than *Hoploscaphites* sp. (20–30 °C), which may have been adapted to mid-to-lower water column and nektobenthic habitats (Landman et al., 2018a; Tsujita and Westermann, 1998). Potentially, the temperature difference between the two ammonite species could indicate thermal stratification of the Late Cretaceous Western Interior Seaway with warmer surface water and cooler temperatures at depth. More specimens are required for definitive statements about thermal stratification to be made.

There are multiple explanations for the temperature difference between the two different bivalve species. In terms of habitats, inoceramids are typically found lying parallel to bedding, which suggests they did not burrow and lived at the sediment-water interface (MacLeod and Hoppe, 1992). Lucinid bivalves were infaunal burrowers that lived in the sediment with high organic carbon content (Taylor et al., 2014), where temperature may be slightly elevated compared to the bottom water–sediment interface. Modern lucinid bivalves are known to host sulfur-oxidizing bacterial symbionts and burrow to high sulfur concentrations in sediments (Bryant et al., 2023). Neither diagenesis nor vital effects can explain the difference in the bivalve samples. Evaluating so-called "vital effects" is important when considering whether biogenic carbonate precipitated in isotopic equilibrium. Although empirical studies of how vital effects may impact triple oxygen isotope values have not yet been done, we find it unlikely that a vital effect would result in an equilibrium triple oxygen isotope value, as seen in the bivalve fossils in this study (Fig. 4). Theoretical work has shown that vital effects caused through the (de) hydroxylation of bicarbonate impose a kinetic, non-equilibrium fractionation effect and would result in the shell carbonate having either lower $\Delta'^{17}O$ and $\delta^{18}O$ values (in the case of hydroxylation) or higher $\Delta'^{17}O$ and $\delta^{18}O$ values (in the case of dehydroxylation) than the equilibrium carbonate value (Fig. 4E; Guo and Zhou, 2019). Similarly, diagenesis would not result in a carbonate that has a $\Delta'^{17}O$ and $\delta^{18}O$ value that plots within our carbonate-water equilibrium field (as shown by the fluid-rock mixing model). Since the fossil bivalve triple oxygen isotope values fall within the equilibrium carbonate-water field (Fig. 4), and until more systematic studies have been performed, we assume that vital effects or



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diagenesis did not impact these samples, and the observed temperature difference is due to environmental factors.

For Campanian samples that have measured values for both paired $\delta^{18}O$-$\Delta'^{17}O$ and $\Delta_{47}$ values, the temperatures calculated from $\Delta_{47}$ [$T(\Delta_{47})$] are equal to or higher than those using $\delta^{18}O$-$\Delta'^{17}O$ thermometry [$T(\Delta'^{17}O)$; Tables 3 and 4; Figs. 4–6]. The *Placenticeras meeki* (UNM15503-1) and *Baculites compressus* (UNM15505-1) fossils have reconstructed temperatures that are within error between the two isotope systems [$T(\Delta'^{17}O) = 33$ °C (±10); $T(\Delta_{47}) = 42$ °C (±7) and $T(\Delta'^{17}O) = 22$ °C (±10); $T(\Delta_{47}) = 27$ °C (±6), respectively; Tables 3 and 4]. This suggests that the ~10 °C difference in formation temperatures is real, and the Campanian Western Interior Seaway had thermal heterogeneity.

A $\Delta'^{17}O$-$\Delta_{47}$ equilibrium fractionation line was calculated using Equations 4 and 11 (black line, Fig. 6). Samples that formed in equilibrium with seawater should have $\Delta_{47}$ and $\Delta'^{17}O$ values that correspond to the calculated $\Delta'^{17}O$-$\Delta_{47}$ equilibrium line for a given temperature. Note, however, that using the $\Delta'^{17}O$-$\Delta_{47}$ equilibrium line for temperature estimates results in large errors. Therefore, we only used the $\Delta'^{17}O$-$\Delta_{47}$ equilibrium line to look at how the two values relate to one another and explore processes that may have impacted the observed values and did not derive any temperature information from the $\Delta'^{17}O$-$\Delta_{47}$ equilibrium line. Campanian fossil shells UNM15503-1 and UNM15505-1 have $\Delta'^{17}O$ and $\Delta_{47}$ values that plot within measurement error (interpreted by the error bars) on the $\Delta'^{17}O$-$\Delta_{47}$ equilibrium fractionation line. Void-filling and crack-filling sparry calcite samples (UNM15502-2 and UNM15505-2) also have similar $\Delta'^{17}O$-$\Delta_{47}$ temperatures (Fig. 6), which we interpret to reflect fluid-rock interaction that affected $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ values in such a way that maintained isotopic equilibrium at the diagenetic conditions. The $T(\Delta_{47})$ values are within the range of fossil-reconstructed $T(\Delta_{47})$ (42 °C and 35 °C for UNM15502-2 and UNM15505-2, respectively; Table 3). However, these samples clearly do not reflect the marine conditions suggested by their $\Delta'^{17}O$-$\delta^{18}O$ values (Fig. 4). Therefore, we interpret this temperature to reflect a minimum temperature of alteration, which indicates that these concretions may have formed rapidly after burial, preserving near-marine conditions as suggested by previous studies (Landman and Klofak, 2012; Maples, 1986; Zatoń and Marynowski, 2006).

### *Maastrichtian Concretions*

One *Hoploscaphites* sp. fossil (UNM15526-1) has $\Delta'^{17}O$ and $\delta^{18}O$ values that correspond to a formation temperature of 32 °C (±10 °C; Fig. 4B) in waters of ~1.5‰ (±2‰). Another *Hoploscaphites* sp. fossil (AMNH161248-4) has the same temperature of formation but suggests formation in 2.6‰ (±2‰) water, outside of our previously defined field of unaltered values. The ammonite $T(\Delta_{47})$ in concretion AMNH161248 is 35 °C, indistinguishable from 32 °C using $\Delta'^{17}O$-$\delta^{18}O$ thermometry. The AMNH161248 concretion also has void-filling sparry calcite that does not plot within the ice-free $\delta^{18}O$ range of seawater (Fig. 4B). The $\Delta'^{17}O$ value of the sparry infill is considerably lower than that of the fossil, which suggests that a late-stage diagenetic fluid interacted with the shell at lower temperature than the fossil growth conditions. The $\Delta_{47}$ value also suggests a lower temperature of formation for the void-filling sparry calcite (23 °C) than the fossil shell (35 °C).

The remaining fossil samples from the nearshore basal Fox Hills Formation of the *Hoploscaphites birklundae* zone in Red Bird, Wyoming (UNM 15525) and the offshore Elk Butte Member of the Pierre Shale of the *Hoploscaphites nicolletii* zone in South Dakota (AMNH 46620) did not plot within the range of an ice-free seawater, which indicates that some alteration has occurred. A highly silicified concretion (UNM15525-1) had the highest $\Delta'^{17}O$ and lowest $\delta^{18}O$ values, which indicates that it was clearly altered. The $T(\Delta_{47})$ reconstruction supports this conclusion. The sample yielded a low calculated water $\delta^{18}O$ value (−7.9‰) and warm (albeit within the range of reported temperatures from this study) $T(\Delta_{47})$ of 38 °C (Table 3).

### *"Seeing through" Diagenesis from Altered Fossil Carbonate*

We used the multiple carbonate phases (fossil aragonite and crack-filling and void-filling sparry calcite) from this study to constrain the diagenetic history of the carbonate concretions using a fluid-rock mixing model in a PSO algorithm (see Methods and Results sections for more information). Histograms of the PSO output show the range of alteration temperatures, meteoric water values, and formation temperatures that best describe the dataset from this study (Figs. 7 and 8). Most optimized alteration temperatures range between 45 °C and 70 °C. However, the optimized formation temperatures occupy a narrow range based on the dataset used. When considering the entire dataset, most of the PSO results suggest a formation temperature of 25 °C (ranging between 22 °C and 25 °C; Fig. 7). When the dataset is broken up into time periods and samples that have higher or lower $\Delta'^{17}O$ values, the formation temperatures change (Fig. 8). The Maastrichtian samples have formation temperatures that range from 9 °C to 27 °C. The Campanian samples have formation temperatures that range between 25 °C and 38 °C. This suggests a potential cooling of the ocean during the Maastrichtian, similar (albeit a larger total change) to global sea-surface temperature changes interpreted from $TEX_{86}$ and benthic foraminifera $\delta^{18}O$ global records (Huber et al., 2018; O'Brien et al., 2017).

We assume that the diagenetic fluids that altered the carbonate are meteoric in origin. Therefore, the PSO output for the alteration fluid $\delta^{18}O$ value is ultimately set by paleo-precipitation $\delta^{18}O$ values. We find that paleo-precipitation $\delta^{18}O$ values between −12‰ and −9‰ best describe all of the data (Fig. 7). The optimized $\delta^{18}O$ value is lower when the alteration temperature is also lower and higher as the alteration temperature increases closer to 75 °C (Figs. S3–S6). When analyzing the dataset by the high- and low-temperature groups within each time period, the high-temperature Campanian group had alteration fluid $\delta^{18}O$ values lower than the average dataset (−14‰ to −11‰), and the low-temperature Maastrichtian group had alteration fluid $\delta^{18}O$ values higher than the average dataset (−10‰ to −7‰; Fig. 8). Although different from each other, these results still overlap with the average $\delta^{18}O$ value of the alteration fluid from the entire dataset. The paleo-precipitation $\delta^{18}O$ value of −12‰ to −9‰ is slightly higher than the −20‰ to −14‰ proposed in Fricke et al. (2010) to represent high-elevation recharge during summer monsoons. However, the low $\delta^{18}O$ value of −12‰ to −9‰ must represent some extent of paleo-elevation as proposed in Fricke et al. (2010) and Poulsen et al. (2007), since precipitation sourced through evaporation of nearby seawater without significant elevation change would have considerably higher $\delta^{18}O$ values. Therefore, to achieve paleo-precipitation $\delta^{18}O$ values of −12‰ to −9‰ requires the infiltration of fluids sourced from higher elevations into the shallow sediments at the time of early carbonate diagenesis.

## Implications of Interpreting Habitat Environments Using Fossil Shell $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ Values without Correcting for Any Diagenesis

Overall, if the $\Delta_{47}$ values are taken as primary, there is no difference in temperature or $\delta^{18}O_{seawater}$ values between the two periods, and the positive Western Interior Seaway $\delta^{18}O_{seawater}$ value was a result of evaporation, which increases the $\delta^{18}O$ value of a water body. The $\Delta_{47}$ values of the fossil carbonate reflect $\delta^{18}O_{seawater}$ values of >0‰ and temperatures ≥35 °C, except for AMNH46620-1 [$T(\Delta_{47}) = 31$ °C; offshore,



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Maastrichtian sample] and UNM15505-1 [T($\Delta_{47}$) = 27 °C; non-seep, Campanian sample; Fig. 5]. The highest T($\Delta_{47}$) is 42 °C from near-seep Campanian sample UNM15503-1 (Fig. 5).

The $\Delta'^{17}$O and $\delta^{18}$O values of the fossil carbonate reflect a similar temperature history as the $\Delta_{47}$ values but the opposite of the Western Interior Seaway $\delta^{18}O_{seawater}$ values. The Western Interior Seaway $\delta^{18}O_{seawater}$ value must be between −7‰ and +3‰ for all fossil carbonate in this study to reflect initial precipitation environments (Figs. 4C and 4D). Similar to the $\Delta_{47}$ values, there is no difference in the range of $\delta^{18}O_{seawater}$ values between either the Campanian or Maastrichtian samples. The two samples that reflect low $\delta^{18}O_{seawater}$ values (UNM15473 and AMNH46620) also correspond to the lowest temperatures (≤10 °C; Figs. 4C and 4D). The Maastrichtian samples that reflect higher $\delta^{18}O_{seawater}$ values also reflect higher temperatures (~40 °C; Fig. 4D). Although the $\delta^{18}O_{seawater}$ range is not too different from other reported values (Petersen et al., 2016), a $\delta^{18}O_{seawater}$ value of −7‰ is unlikely. The Western Interior Seaway would require an input of 50% freshwater with a $\delta^{18}$O value of −12‰ to change the $\delta^{18}O_{seawater}$ value from −1‰ to −7‰ (Figs. 4C and 4D). Such a flux of freshwater would require ammonites and other cephalopods living in these regions to be insensitive to a 50% drop in salinity, which challenges the existing paleobiological paradigm for these taxa. Additionally, the only <−7‰ $\delta^{18}O_{seawater}$ value derived from the $\Delta_{47}$ values was from a void-filling calcite sample, UNM15505-2 (Fig. 5), which is thought to represent an alteration fluid and not seawater. Based on the triple oxygen isotope values of the fossil, we cannot interpret all fossil data from this study to reflect original habitat or environmental conditions and conclude that diagenesis impacted the isotopic composition of some of the fossils enough to represent sedimentary burial environments.

## Resolving the $\Delta_{47}$ and $\Delta'^{17}$O Values of This Study

For our study, the paired $\Delta'^{17}$O-$\Delta_{47}$ data further validate the samples that were deemed pristine based on their $\delta^{18}$O-$\Delta'^{17}$O values plotting within the equilibrium field from Figure 4 (samples UNM15505-1 and UNM15503-1), and the ~10 °C difference in formation temperature between the two samples is robust. Two samples (UNM15526-1 and AMNH161248-4) have $\Delta'^{17}$O-$\Delta_{47}$ values that are within error of the $\Delta'^{17}$O-$\Delta_{47}$ equilibrium line but do not plot within the triple oxygen isotope equilibrium field. We interpret this as further support of evaporation affecting the Western Interior Seaway seawater $\delta^{18}$O value, which results in a higher $\delta^{18}O_{seawater}$ value during the Maastrichtian. Reconstructed $\delta^{18}O_{seawater}$ values for UNM15526-1 and AMNH161248-4 are 2.6‰ (±1.5‰) and 3.1‰ (±1.2‰) using $\Delta_{47}$ values, and 1.7‰ (±2‰) and 2.8‰ (±2‰) using $\delta^{18}$O-$\Delta'^{17}$O values, respectively. A Campanian fossil (UNM15502-1) has $\delta^{18}$O-$\Delta'^{17}$O values that fall within the triple oxygen equilibrium field but does not appear be in $\Delta'^{17}$O-$\Delta_{47}$ equilibrium (Fig. 6). We are confident that, based on the triple oxygen isotope values, this sample is pristine and reflects formation temperatures and $\delta^{18}O_{seawater}$ values similar to those of other pristine samples from the Campanian dataset (UNM15502-4, UNM15504-2, and UNM15505-1; Figs. 4B and 4D). Therefore, this sample may have $\Delta_{47}$ values that have been altered by a process that does not impact the $\delta^{18}$O or $\Delta'^{17}$O values, mainly a process that involved little fluid interaction such that $\delta^{18}$O and $\Delta'^{17}$O were unaffected. The T($\Delta_{47}$) and $\delta^{18}O_{seawater}$ values are reasonable (37 °C and 1.3‰, respectively); however, the $\delta^{18}$O-$\Delta'^{17}$O reconstructed temperature and $\delta^{18}O_{seawater}$ (23 °C and 1.3‰, respectively) reveals that the sample actually reflects the low-temperature group of the Campanian fossils. This demonstrates the usefulness of paired $\Delta'^{17}$O-$\Delta_{47}$ values to distinguish small differences within a dataset. Fossils AMNH46620-1 and UNM15473-1 and infill sample AMNH161248-1 all have $\Delta'^{17}$O-$\Delta_{47}$ values that are not in equilibrium, similar to Figure 4, which shows $\delta^{18}$O-$\Delta'^{17}$O values.

It is possible, but not presently well known, whether vital effects could impact the $\Delta'^{17}$O, $\delta^{18}$O, and $\Delta_{47}$ values differently. Metabolic vital effects have been shown to impact $\delta^{13}$C values of fast-growing ammonites (McConnaughey et al., 1997; Tobin and Ward, 2015). Clumped isotope measurements in high shell growth areas of modern cephalopods, *Nautilus* sp., have yielded lower $\Delta_{47}$ values—i.e., higher T($\Delta_{47}$)—than are predicted from ambient seawater temperature, though the available studies were hampered by legacy clumped isotope data correction issues (e.g., Dennis et al., 2013) or poorly known thermal histories of the specimens studied (i.e., Nautilus are known to make large diurnal migrations from cooler deep waters to the warmer surface, and back; Davies et al., 2021; Dunstan et al., 2011). Cephalopod aragonite may be an example of negligible $\delta^{18}$O vital effect offsets and potentially important $\Delta_{47}$ vital effect offsets (Davies et al., 2021; Dennis et al., 2013). Additional research is needed to fully evaluate whether $\Delta_{47}$ values that appear too warm in fossil taxa are related to these vital effects, burial alteration, or some combination of both.

As described earlier when discussing vital effects in bivalves, it is unknown how vital effects impact triple oxygen isotope values. If the vital effect causes non-equilibrium precipitation due to the hydroxylation or dehydroxylation of bicarbonate (through changes in $CO_2$), then the carbonate triple oxygen isotope composition is expected to deviate from expected equilibrium with either lower $\Delta'^{17}$O and $\delta^{18}$O (for hydroxylation) or higher $\Delta'^{17}$O and $\delta^{18}$O values (for dehydroxylation; Fig. 4E). We cannot attribute vital effects impacting the triple oxygen isotope values of biogenic carbonate that plot within the triple oxygen equilibrium field (Fig. 4) because we do not, as of yet, have a vital effect mechanism that would cause equilibrium precipitation. Therefore, we interpret those oxygen isotope values as primary and reflecting initial precipitation conditions. Minor hydroxylation disequilibrium vital effects overlap minor diagenetic effects (Fig. 4E). Therefore, we cannot distinguish between vital effects or meteoric diagenesis for samples UNM15502-1, UNM15473-1, and AMNH46620-1. However, combined with the void-filling and crack-filling sparry calcite samples that we know formed through meteoric water infiltration, "seeing through" diagenesis still adequately describes those samples.

A clear observation from all of our data is that fossil shell clumped isotope temperatures are mostly higher (albeit usually within error) than those from triple oxygen isotopes (Figs. 4 and 5). However, many of the fossil shell $\delta^{18}$O and $\Delta'^{17}$O values that plot within the seawater equilibrium field (Fig. 4) have $\Delta'^{17}$O-$\Delta_{47}$ temperatures that plot within measurement error (error bars) of the $\Delta'^{17}$O-$\Delta_{47}$ equilibrium line (Fig. 6). Samples that have $\Delta'^{17}$O and $\Delta_{47}$ values that plot on the $\Delta'^{17}$O-$\Delta_{47}$ equilibrium line but do not have $\Delta'^{17}$O-$\delta^{18}$O values that plot within the carbonate-water equilibrium field (namely, the void- and crack-filling sparry calcite samples UNM15502-2 and UNM15505-2), reflect re-equilibration at high fluid-rock ratios impacting the $\Delta'^{17}$O, $\delta^{18}$O, and $\Delta_{47}$ values and are not a reflection of primary temperatures.

The analytical challenge of triple oxygen and clumped isotope analysis lies in the fact that they are infrequently measured on the same samples. Future work focusing on the co-analysis of the two systems in a variety of carbonate-forming settings may or may not illustrate their co-dependence. The $\Delta'^{17}$O-$\Delta_{47}$ equilibrium line combines two independent variables as an additional check on diagenesis. This could be particularly useful in samples where the reconstructed clumped isotope temperature is either slightly higher or slightly lower than expected, and it is unclear whether some other process may have impacted the isotopic values (as is the case for samples UNM15502-1, UNM15473-1, and AMNH46620-1; Fig. 6).



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### Implications for Oceanographic Reconstructions of the Late Cretaceous Western Interior Seaway

Temperature (20–40 °C) and $\delta^{18}O_{seawater}$ (−3‰–0‰) values from the triple oxygen isotopes and most of the clumped isotope measurements in this study overlap with clumped isotope temperatures for the Campanian derived by previous carbonate analyses from the Western Interior Seaway (Fig. 9; Dennis et al., 2013; Gao et al., 2021; Petersen et al., 2016). For the Maastrichtian Western Interior Seaway, our temperatures are similar to, albeit cooler than, those of Petersen et al. (2016) and Dennis et al. (2013; 10–25 °C, this study, versus ~20 °C, previous studies). However, the Western Interior Seaway $\delta^{18}O_{seawater}$ values reported in this study are slightly higher than those reported in Petersen et al. (2016) and Dennis et al. (2013). We do not think that this represents a non-homogeneous Western Interior Seaway during the Maastrichtian, as suggested in Petersen et al. (2016), because ~50% of the seawater would need to be freshwater to change the Western Interior Seaway $\delta^{18}O_{seawater}$ value to −6‰ and would drastically change the salinity of the Western Interior Seaway and habitability for marine organisms (Figs. 4 and 9). Alternatively, the potentially shallower Maastrichtian Western Interior Seaway would result in higher $\delta^{18}O_{seawater}$ values due to evaporation, as suggested by the triple oxygen isotope values of the Maastrichtian fossils. An expansion of the Maastrichtian datasets with the inclusion of careful $\Delta'^{17}O$ and $\Delta_{47}$ analyses is necessary to determine the difference in connectedness of the Western Interior Seaway to the open ocean during the early and Late Maastrichtian.

## CONCLUSIONS

Through this study, we show that primary paleoenvironmental information can be deconvolved from even minor diagenetic effects by combining $\Delta'^{17}O$, $\delta^{18}O$, and $\Delta_{47}$ measurements of fossil carbonate. Additionally, we can "see through" diagenesis by measuring additional carbonate phases, such as void-filling or crack-filling calcite, and estimate initial precipitation environments in altered fossil carbonates. This combined measurement approach allows for better characterization of paleoenvironmental conditions in the Late Cretaceous North American Western Interior Seaway. Our results are not only useful for paleoceanographic reconstructions, but also investigations of environment-life interactions in an important marine system leading into a major mass extinction.

Based on the data from this study, we make the following conclusions regarding the Late Cretaceous Western Interior Seaway:

(1) Average temperatures of the Western Interior Seaway were between 20 °C and 40 °C. Temperatures from fossil shells are 10–40 °C from triple oxygen isotope constraints and 26–42 °C from clumped isotopes. Results of the PSO algorithm suggest that the temperature of the Maastrichtian Western Interior Seaway was 10–15 °C cooler than during the Campanian.

(2) Maastrichtian seawater may have been restricted and affected by more evaporation (resulting in a higher $\delta^{18}O$ value) than Campanian seawater (which has $\delta^{18}O$ values similar to those expected of the open ocean in a primarily ice-free Cretaceous world).

(3) The Western Interior Seaway during the Campanian may have been thermally stratified, with ~10 °C warmer surface water and cooler bottom water. This is supported by both $\Delta_{47}$ and $\delta^{18}O$-$\Delta'^{17}O$ values and PSO algorithm results.

(4) Diagenesis occurred via the infiltration of meteoric water into the carbonate concretions. All of the void-filling and crack-filling calcite (in the form of sparry infill) in the concretions interacted with a light meteoric water (~−10‰), which suggests moderate topographic relief on the western coast of the Western Interior Seaway or that groundwater represented a mixture of high- and low-altitude precipitation (Fricke et al., 2010; Poulsen et al., 2007). We confirm that the sparry calcite from these concretions are not a good proxy for determining primary Cretaceous oceanic temperatures, but does provide important information on burial conditions and the diagenetic history of the region.

(5) Both the PSO and paired $\Delta'^{17}O$-$\Delta_{47}$ values support that the carbonate concretions from this dataset formed and were altered rapidly after shallow burial in temperatures of ~40 °C.

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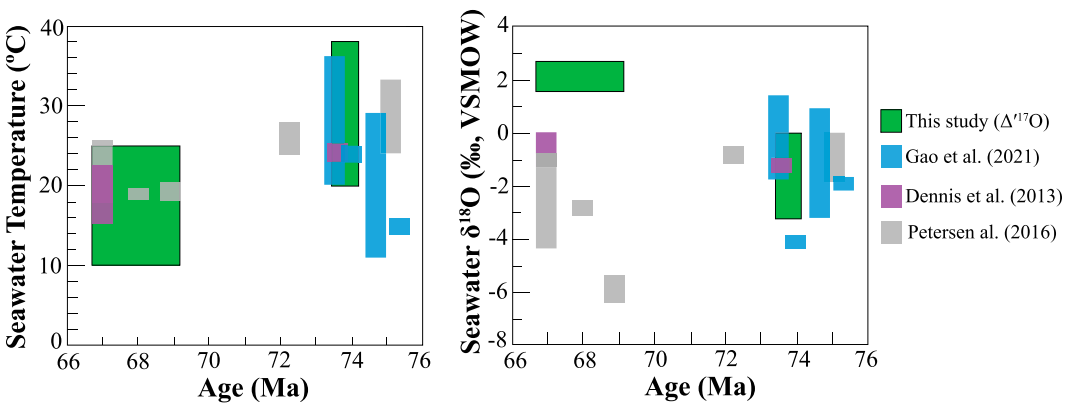


**Figure 9. Comparison of triple oxygen isotope temperature and $\delta^{18}O_{seawater}$ results from this study to previous clumped isotope temperatures and $\delta^{18}O_{seawater}$ values from the Western Interior Seaway. Shaded areas represent the range of published values, not including reported error. (A) Temperatures between 20 °C and 40 °C for the Late Campanian (75–72 Ma) Western Interior Seaway from this study (green) are within the range of previous studies Gao et al. (2021, blue), Dennis et al. (2013, purple), and Petersen et al. (2016, gray). The early Maastrichtian (69–67 Ma) temperature range is larger, but overlapping, previous studies from Dennis et al. (2013) and Petersen et al. (2016). (B) This study reports $\delta^{18}O$ values of between −3‰ and 2‰ and agrees with other reported $\delta^{18}O$ values. However, this study suggests that carbonate clumped isotope measurements with calculated $\delta^{18}O$ values of water that are less than −3‰ are an artifact of diagenesis and have reset carbonate clumped isotope values. Note that Petersen et al. (2016) was corrected to the new calibration presented in Jones et al. (2022). VSMOW—Vienna standard mean ocean water.**



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## REFERENCES CITED

Aron, P.G., Levin, N.E., Beverly, E.J., Huth, T.E., Passey, B.H., Pelletier, E.M., Poulsen, C.J., Winkelstern, I.Z., and Yarian, D.A., 2021, Triple oxygen isotopes in the water cycle: Chemical Geology, v. 565, https://doi.org/10.1016/j.chemgeo.2020.120026.

Arthur, M.A., Dean, W.E., and Schlanger, S.O., 1985, Variations in the global carbon cycle during the Cretaceous related to climate, volcanism, and changes in atmospheric $CO_2$, *in* Sundquist, E.T., and Broecker, W.S., eds., The Carbon Cycle and Atmospheric $CO_2$: Natural Variations Archean to Present: American Geophysical Union Geophysical Monograph 32, p. 504–529, https://doi.org/10.1029/GM032p0504.

Banner, J.L., and Hanson, G.N., 1990, Calculation of simultaneous isotopic and trace element variations during water-rock interaction with applications to carbonate diagenesis: Geochimica et Cosmochimica Acta, v. 54, p. 3123–3137, https://doi.org/10.1016/0016-7037(90)90128-8.

Barrera, E., and Savin, S.M., 1999, Evolution of Late Campanian–Maastrichtian marine climates and oceans, *in* Barrera, E., and Johnson, C.C., Evolution of the Cretaceous Ocean-Climate System: Geological Society of America Special Paper 332, p. 245–282, https://doi.org/10.1130/0-8137-2332-9.245.

Bernasconi, S.M., et al., 2021, InterCarb: A community effort to improve interlaboratory standardization of the carbonate clumped isotope thermometer using carbonate standards: Geochemistry, Geophysics, Geosystems, v. 22, https://doi.org/10.1029/2020GC009588.

Brusatte, S.L., Butler, R.J., Barrett, P.M., Carrano, M.T., Evans, D.C., Lloyd, G.T., Mannion, P.D., Norell, M.A., Peppe, D.J., Upchurch, P., and Williamson, T.E., 2015, The extinction of the dinosaurs: Biological Reviews of the Cambridge Philosophical Society, v. 90, p. 628–642, https://doi.org/10.1111/brv.12128.

Bryant, R.N., Richardson, J.A., Kalia, T.C., Gros, O., Lopez-Garriga, J., and Blättler, C.L., 2023, Inorganic sulfate–based signatures of chemosymbiosis in modern infaunal lucinids: Geology, v. 51, p. 1022–1026, https://doi.org/10.1130/G51353.1.

Clarke, L.J., and Jenkyns, H.C., 1999, New oxygen isotope evidence for long-term Cretaceous climatic change in the Southern Hemisphere: Geology, v. 27, p. 699–702, https://doi.org/10.1130/0091-7613(1999)027<0699:NOIEFL>2.3.CO;2.

Cobban, W.A., Mereweather, E.A., Fouch, T.D., and Obradovich, J.D., 1994, Some Cretaceous shorelines in the Western Interior of the United States, *in* Caputo, M.V., Peterson, J.A., and Franczyk, K.J., eds., Mesozoic Systems of the Rocky Mountain Region: Denver, Society for Sedimentary Geology (SEPM) Rocky Mountain Section.

Cobban, W.A., Walaszczyk, I., Obradovich, J.D., and McKinney, K.C., 2006, A USGS zonal table for the Upper Cretaceous middle Cenomanian–Maastrichtian of the Western Inerior of the United States based on ammonites, inoceramids, and radiometric ages: U.S Geological Survey Open-File Report 2006-1250, 45 p.

Cochran, J.K., Landman, N.H., Turekian, K.K., Michard, A., and Schrag, D.P., 2003, Paleoceanography of the Late Cretaceous (Maastrichtian) Western Interior Seaway of North America: Evidence from Sr and O isotopes: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 191, p. 45–64, https://doi.org/10.1016/S0031-0182(02)00642-9.

Cochran, J.K., Kallenberg, K., Landman, N.H., Harries, P.J., Weinreb, D., Turekian, K.K., Beck, A.J., and Cobban, W.A., 2010, Effect of diagenesis on the Sr, O, and C isotope composition of Late Cretaceous mollusks from the Western Interior Seaway of North America: American Journal of Science, v. 310, p. 69–88, https://doi.org/10.2475/02.2010.01.

Condamine, F.L., Guinot, G., Benton, M.J., and Currie, P.J., 2021, Dinosaur biodiversity declined well before the asteroid impact, influenced by ecological and environmental pressures: Nature Communications, v. 12, 3833, https://doi.org/10.1038/s41467-021-23754-0.

Coulson, A.B., Kohn, M.J., and Barrick, R.E., 2011, Isotopic evaluation of ocean circulation in the Late Cretaceous North American seaway: Nature Geoscience, v. 4, p. 852–855, https://doi.org/10.1038/ngeo1312.

Craig, H., and Gordon, L.I., 1965, Deuterium and oxygen 18 variations in the ocean and the marine atmosphere, *in* Tongiorgi, E., ed., Stable Isotopes in Oceanographic Studies and Paleotemperatures: Spoleto, p. 9–130.

Davies, A.J., Davis, S., and John, C.M., 2021, Evidence of taxonomic non-equilibrium effects in the clumped isotope composition of modern cephalopod carbonate: Chemical Geology, v. 578, https://doi.org/10.1016/j.chemgeo.2021.120317.

Dennis, K.J., and Schrag, D.P., 2010, Clumped isotope thermometry of carbonatites as an indicator of diagenetic alteration: Geochimica et Cosmochimica Acta, v. 74, p. 4110–4122, https://doi.org/10.1016/j.gca.2010.04.005.

Dennis, K.J., Affek, H.P., Passey, B.H., Schrag, D.P., and Eiler, J.M., 2011, Defining an absolute reference frame for 'clumped' isotope studies of $CO_2$: Geochimica et Cosmochimica Acta, v. 75, p. 7117–7131, https://doi.org/10.1016/j.gca.2011.09.025.

Dennis, K.J., Cochran, J.K., Landman, N.H., and Schrag, D.P., 2013, The climate of the Late Cretaceous: New insights from the application of the carbonate clumped isotope thermometer to Western Interior Seaway macrofossil: Earth and Planetary Science Letters, v. 362, p. 51–65, https://doi.org/10.1016/j.epsl.2012.11.036.

Dunstan, A.J., Ward, P.D., and Marshall, N.J., 2011, Vertical distribution and migration patterns of Nautilus pompilius: PLoS One, v. 6, https://doi.org/10.1371/journal.pone.0016311.

Eiler, J.M., 2011, Paleoclimate reconstruction using carbonate clumped isotope thermometry: Quaternary Science Reviews, v. 30, p. 3575–3588, https://doi.org/10.1016/j.quascirev.2011.09.001.

Fricke, H.C., Foreman, B.Z., and Sewall, J.O., 2010, Integrated climate model-oxygen isotope evidence for a North American monsoon during the Late Cretaceous: Earth and Planetary Science Letters, v. 289, p. 11–21, https://doi.org/10.1016/j.epsl.2009.10.018.

Gao, Y., Henkes, G.A., Cochran, J.K., and Landman, N.H., 2021, Temperatures of Late Cretaceous (Campanian) methane-derived authigenic carbonates from the Western Interior Seaway, South Dakota, USA, using clumped isotopes: Geological Society of America Bulletin, v. 133, p. 2524–2534, https://doi.org/10.1130/B35846.1.

Geraghty, M.D., and Westermann, G.E.G., 1994, Origin of Jurassic ammonite concretions assemblages at Alfeld, Germany: A biogenic alternative: Palaontologische Zeitschrift, v. 68, p. 473–490, https://doi.org/10.1007/BF02991357.

Gill, J.R., and Cobban, W.A., 1966, The Red Bird section of the Upper Cretaceous Pierre Shale of Wyoming: U.S Geological Survey Professional Paper 393-A, 73 p.

Guo, W., and Zhou, C., 2019, Triple oxygen isotope fractionation in the DIC-$H_2O$-$CO_2$ system: A numerical framework and its implications: Geochimica et Cosmochimica Acta, v. 246, p. 541–564, https://doi.org/10.1016/j.gca.2018.11.018.

Hay, W., Eicher, D., and Diner, R., 1993, Physical oceanography and water masses of the Cretaceous Western Interior Seaway, *in* Caldwell, W.G.E., and Kauffman, E.G., eds., Evolution of the Western Interior Basin: Geological Association of Canada Special Paper 39, p. 297–318.

Hay, W.W., 2008, Evolving ideas about the Cretaceous climate and ocean circulation: Cretaceous Research, v. 29, p. 725–753, https://doi.org/10.1016/j.cretres.2008.05.025.

Hayles, J., Gao, C., Cao, X., Liu, Y., and Bao, H., 2018, Theoretical calibration of the triple oxygen isotope thermometer: Geochimica et Cosmochimica Acta, v. 235, p. 237–245, https://doi.org/10.1016/j.gca.2018.05.032.

He, S., Kyser, T.K., and Caldwell, W.G.E., 2005, Paleoenvironment of the Western Interior Seaway inferred from $\delta^{18}O$ and $\delta^{13}C$ values of molluscs from the Cretaceous Bearpaw marine cyclothem: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 217, p. 67–85.

Hemingway, J.D., and Henkes, G.A., 2021, A disordered kinetic model for clumped isotope bond reordering in carbonates: Earth and Planetary Science Letters, v. 566, https://doi.org/10.1016/j.epsl.2021.116962.

Henkes, G.A., Passey, B.H., Grossman, E.L., Shenton, B.J., Pérez-Huerta, A., and Yancey, T.E., 2014, Temperature limits for preservation of primary calcite clumped isotope paleotemperatures: Geochimica et Cosmochimica Acta, v. 139, p. 362–382, https://doi.org/10.1016/j.gca.2014.04.040.

Hicks, J.F., Obradovich, J.D., and Tauxe, L., 1999, Magnetostratigraphy, isotopic age calibration and intercontinental correlation of the Red Bird section of the Pierre Shale, Niobrara County, Wyoming, USA: Cretaceous Research, v. 20, p. 1–27, https://doi.org/10.1006/cres.1998.0133.

Huber, B.T., Hodell, D.A., and Hamilton, C.P., 1995, Middle–Late Cretaceous climate of the southern high latitudes: Stable isotopic evidence for minimal equator-to-pole thermal gradients: Geological Society of America Bulletin, v. 107, p. 1164–1191, https://doi.org/10.1130/0016-7606(1995)107<1164:MLCCOT>2.3.CO;2.

Huber, B.T., Norris, R.D., and MacLeod, K.G., 2002, Deep-sea paleotemperature record of extreme warmth during the Cretaceous: Geology, v. 30, p. 123–126, https://doi.org/10.1130/0091-7613(2002)030<0123:DSPROE>2.0.CO;2.

Huber, B.T., MacLeod, K.G., Watkins, D.K., and Coffin, M.F., 2018, The rise and fall of the Cretaceous Hot Greenhouse climate: Global and Planetary Change, v. 167, p. 1–23, https://doi.org/10.1016/j.gloplacha.2018.04.004.

Hulston, J.R., and Thode, H.G., 1965, Variations in the $S^{33}$, $S^{34}$, and $S^{36}$ contents of meteorites and their relation to chemical and nuclear effects: Journal of Geophysical Research, v. 70, p. 3475–3484, https://doi.org/10.1029/JZ070i014p03475.

Huntington, K.W., and Petersen, S.V., 2023, Frontiers of carbonate clumped isotope thermometry: Annual Review of Earth and Planetary Sciences, v. 51, p. 611–641, https://doi.org/10.1146/annurev-earth-031621-085949.

Huth, T.E., Passey, B.H., Cole, J.E., Lachniet, M.S., McGee, D., Denniston, R.F., Truebe, S., and Levin, N.E., 2022, A framework for triple oxygen isotopes in speleothem paleoclimatology: Geochimica et Cosmochimica Acta, v. 319, p. 191–219, https://doi.org/10.1016/j.gca.2021.11.002.

Jones, M.M., Petersen, S.V., and Curley, A.N., 2022, A tropically hot mid-Cretaceous North American Western Interior Seaway: Geology, v. 50, p. 954–958, https://doi.org/10.1130/G49998.1.

Kauffman, E., 1984, Paleobiogeography and evolutionary response dynamic in the Cretaceous Western Interior Seaway of North America, *in* Westermann, G.E.G., ed., Jurassic–Cretaceous Biochronology and Paleogeography of North America: Geological Association of Canada Special Paper 27, p. 273–306.

Kauffman, E., and Caldwell, W., 1993, The Western Interior Basin in Space and Time: Geological Association of Canada Special Paper 39, p. 1–30.

Kelson, J.R., Petersen, S.V., Niemi, N.A., Passey, B.H., and Curley, A.N., 2022, Looking upstream with clumped and triple oxygen isotopes of estuarine oyster shells in the early Eocene of California, USA: Geology, v. 50, p. 755–759, https://doi.org/10.1130/G49634.1.

Kennedy, W.J., Landman, N.H., Christensen, W.K., Cobban, W.A., and Hancock, J.M., 1998, Marine connections in North America during the Late Maastrichtian: Palaeogeographic and palaeobiogeographic significance of Jeletzkytes nebrascensisZone cephalopod fauna from the Elk Butte Member of the Pierre Shale, SE South Dakota and NE Nebraska: Cretaceous Research, v. 19, p. 745–775, https://doi.org/10.1006/cres.1998.0129.

Kim, S.-T., Coplen, T.B., and Horita, J., 2015, Normalization of stable isotope data for carbonate minerals: Implementation of IUPAC guidelines: Geochimica et Cosmochimica Acta, v. 158, p. 276–289, https://doi.org/10.1016/j.gca.2015.02.011.



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Knoll, K., Landman, N.H., Cochran, J.K., Macleod, K.G., and Sessa, J.A., 2016, Microstructural preservation and the effects of diagenesis on the carbon and oxygen isotope composition of Late Cretaceous aragonitic mollusks from the Gulf Coastal Plain and the Western Interior Seaway: American Journal of Science, v. 316, p. 591–613, https://doi.org/10.2475/07.2016.01.

Landman, N.H., and Klofak, S.M., 2012, Anatomy of a concretion: Life, death, and burial in the Western Interior Seaway: Palaios, v. 27, p. 671–692, https://doi.org/10.2110/palo.2011.p11-105r.

Landman, N.H., and Waage, K.M., 1993, Scaphitid ammonites of the Upper Cretaceous (Maastrichtian) Fox Hills Formation in South Dakota and Wyoming: Bulletin of the American Museum of Natural History, no. 215.

Landman, N.H., Remin, Z., Garb, M.P., and Chamberlain, J.A., Jr., 2013, Cephalopods from the Badlands National Park area, South Dakota: Reassessment of the position of the Cretaceous/Paleogene boundary: Cretaceous Research, v. 42, p. 1–27, https://doi.org/10.1016/j.cretres.2012.12.011.

Landman, N.H., Goolaerts, S., Jagt, J.W.M., Jagt-Yazykova, E.A., and Machalski, M., 2015, Ammonites on the brink of extinction: Diversity, abundance, and ecology of the order Ammonoidea at the Cretaceous/Paleogene (K/Pg) boundary, *in* Klug, C., Korn, D., De Baets, K., Kruta, I., and Mapes, R.H., eds., Ammonoid Paleobiology: From Macroevolution to Paleogeography: Dordrecht, Netherlands, Springer, p. 497–553, https://doi.org/10.1007/978-94-017-9633-0_19.

Landman, N.H., Cochran, J.K., Slovacek, M., Larson, N.L., Garb, M.P., Brezina, J., and Witts, J.D., 2018a, Isotope sclerochronology of ammonites (Baculites Compressus) from methane seep and non-seep sites in the Late Cretaceous Western Interior Seaway, USA: Implications for ammonite habitat and mode of life: American Journal of Science, v. 318, p. 603–639, https://doi.org/10.2475/06.2018.01.

Landman, N.H., Jicha, B.R., Cochran, J.K., Garb, M.P., Brophy, S.K., Larson, N.L., and Brezina, J., 2018b, $^{40}Ar/^{39}Ar$ date of a bentonite associated with a methane seep deposit in the upper Campanian Baculites compressus Zone, Pierre Shale, South Dakota: Cretaceous Research, v. 90, p. 90–96, https://doi.org/10.1016/j.cretres.2018.03.024.

LeGrande, A.N., and Schmidt, G.A., 2006, Global gridded data set of the oxygen isotopic composition in seawater: Geophysical Research Letters, v. 33, https://doi.org/10.1029/2006GL026011.

Looser, N., Petschnig, P., Hemingway, J.D., Fernandez, A., Morales Grafulha, L., Perez-Huerta, A., Vickers, M.L., Price, G.D., Schmidt, M.W., and Bernasconi, S.M., 2023, Thermally-induced clumped isotope resetting in belemnite and optical calcites: Towards material-specific kinetics: Geochimica et Cosmochimica Acta, v. 350, p. 1–15, https://doi.org/10.1016/j.gca.2023.03.030.

Luz, B., and Barkan, E., 2010, Variations of $^{17}O/^{16}O$ and $^{18}O/^{16}O$ in meteoric waters: Geochimica et Cosmochimica Acta, v. 74, p. 6276–6286, https://doi.org/10.1016/j.gca.2010.08.016.

MacLeod, K.G., and Hoppe, K.A., 1992, Evidence that inoceramid bivalves were benthic and harbored chemosynthetic symbionts: Geology, v. 20, p. 117–120, https://doi.org/10.1130/0091-7613(1992)020<0117:ETIBWB>2.3.CO;2.

Maples, C.G., 1986, Enhanced paleoecological and paleoenvironmental interpretations result from analysis of early diagenetic concretions in Pennsylvanian shales: Palaios, v. 1, p. 512–516, https://doi.org/10.2307/3514633.

McConnaughey, T.A., Burdett, J., Whelan, J.F., and Paull, C.K., 1997, Carbon isotopes in biological carbonates: Respiration and photosynthesis: Geochimica et Cosmochimica Acta, v. 61, p. 611–622, https://doi.org/10.1016/S0016-7037(96)00361-4.

McGunnigle, J.P., Cano, E.J., Sharp, Z.D., Muehlenbachs, K., Cole, D., Hardman, M.F., Stachel, T., and Pearson, D.G., 2022, Triple oxygen isotope evidence for a hot Archean ocean: Geology, v. 50, p. 991–995, https://doi.org/10.1130/G50230.1.

McKinney, C.R., McCrea, J.M., Epstein, S., Allen, H.A., and Urey, H.C., 1950, Improvements in mass spectrometers for the measurement of small differences in isotope abundance ratios: The Review of Scientific Instruments, v. 21, p. 724–730, https://doi.org/10.1063/1.1745698.

Miall, A.D., Catuneanu, O., Vakarelov, B.K., and Post, R., 2008, Chapter 9: The Western Interior Basin, *in* Miall, A.D., ed., Sedimentary Basins of the World: Elsevier, p. 329–362.

Miller, M.F., 2002, Isotopic fractionation and the quantification of $^{17}O$ anomalies in the oxygen three-isotope system: An appraisal and geochemical significance: Geochimica et Cosmochimica Acta, v. 66, p. 1881–1889, https://doi.org/10.1016/S0016-7037(02)00832-3.

O'Brien, C.L., et al., 2017, Cretaceous sea-surface temperature evolution: Constraints from TEX86 and planktonic foraminiferal oxygen isotopes: Earth-Science Reviews, v. 172, p. 224–247, https://doi.org/10.1016/j.earscirev.2017.07.012.

Passey, B.H., and Henkes, G.A., 2012, Carbonate clumped isotope bond reordering and geospeedometry: Earth and Planetary Science Letters, v. 351–352, p. 223–236, https://doi.org/10.1016/j.epsl.2012.07.021.

Petersen, S.V., Tabor, C.R., Lohmann, K.C., Poulsen, C.J., Meyer, K.W., Carpenter, S.J., Erickson, J.M., Matsunaga, K.K.S., Smith, S.Y., and Sheldon, N.D., 2016, Temperature and salinity of the Late Cretaceous Western Interior Seaway: Geology, v. 44, p. 903–906, https://doi.org/10.1130/G38311.1.

Petersen, S.V., et al., 2019, Effects of improved $^{17}O$ correction on interlaboratory agreement in clumped isotope calibrations, estimates of mineral-specific offsets, and temperature dependence of acid digestion fractionation: Geochemistry, Geophysics, Geosystems, v. 20, p. 3495–3519, https://doi.org/10.1029/2018GC008127.

Pörtner, H.O., 2002, Climate variations and the physiological basis of temperature dependent biogeography: Systemic to molecular hierarchy of thermal tolerance in animals: Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology, v. 132, p. 739–761, https://doi.org/10.1016/S1095-6433(02)00045-4.

Poulsen, C.J., Pollard, D., and White, T.S., 2007, General circulation model simulation of the $\delta^{18}O$ content of continental precipitation in the middle Cretaceous: A model-proxy comparison: Geology, v. 35, p. 199–202, https://doi.org/10.1130/G23343A.1.

Raiswell, R., and Fisher, Q.J., 2000, Mudrock-hosted carbonate concretions: A review of growth mechanisms and their influence on chemical and isotopic composition: Journal of the Geological Society, v. 157, p. 239–251, https://doi.org/10.1144/jgs.157.1.239.

Raiswell, R., and Fisher, Q.J., 2004, Rates of carbonate cementation associated with sulphate reduction in DSDP/ODP sediments: Implications for the formation of concretions: Chemical Geology, v. 211, p. 71–85, https://doi.org/10.1016/j.chemgeo.2004.06.020.

Reeside, J.B., and Cobban, W.A., 1960, Studies of the Mowry shale (Cretaceous) and contemporary formations in the United States and Canada: U.S. Geological Survey Professional Paper 355, 126 p., https://doi.org/10.3133/pp355.

Roberts, L.N.R., and Kirschbaum, M.A., 1995, Paleogeography and the Late Cretaceous of the Western Interior of middle North America; coal distribution and sediment accumulation: U.S. Geological Survey Professional Paper 1561, 115 p., https://doi.org/10.3133/pp1561.

Rowe, A., Landman, N., Cochran, J., Witts, J., and Garb, M., 2020, Late Cretaceous methane seeps as habitats for newly hatched ammonites: Palaios, v. 35, p. 151–163, https://doi.org/10.2110/palo.2019.105.

Royer, D.L., Berner, R.A., and Beerling, D.J., 2001, Phanerozoic atmospheric $CO_2$ change: Evaluating geochemical and paleobiological approaches: Earth-Science Reviews, v. 54, p. 349–392, https://doi.org/10.1016/S0012-8252(00)00042-8.

Ryb, U., and Eiler, J.M., 2018, Oxygen isotope composition of the Phanerozoic ocean and a possible solution to the dolomite problem: Proceedings of the National Academy of Sciences of the United States of America, v. 115, p. 6602–6607, https://doi.org/10.1073/pnas.1719681115.

Sachs, J.P., and Schwab, V.F., 2011, Hydrogen isotopes in dinosterol from the Chesapeake Bay estuary: Geochimica et Cosmochimica Acta, v. 75, p. 444–459, https://doi.org/10.1016/j.gca.2010.10.013.

Sakamoto, M., Benton, M.J., and Venditti, C., 2016, Dinosaurs in decline tens of millions of years before their final extinction: Proceedings of the National Academy of Sciences of the United States of America, v. 113, p. 5036–5040, https://doi.org/10.1073/pnas.1521478113.

Schauble, E.A., and Young, E.D., 2021, Mass dependence of equilibrium oxygen isotope fractionation in carbonate, nitrate, oxide, perchlorate, phosphate, silicate, and sulfate minerals: Reviews in Mineralogy and Geochemistry, v. 86, p. 137–178, https://doi.org/10.2138/rmg.2021.86.04.

Schmidt, G., Bigg, G., and Rohling, E., 1999, Global seawater oxygen-18 database: http://data.giss.nasa.gov/o18data.

Shackleton, N.J., and Kennett, J.P., 1975, Paleotemperature history of the Cenozoic and the initiation of Antarctic glaciation: Oxygen and carbon isotope analyses in DSDP sites 277, 279, and 281: Deep Sea Drilling Projects Initial Reports, v. 29, p. 743–755, https://doi.org/10.2973/dsdp.proc.29.117.1975.

Sharp, Z.D., and Wostbrock, J.A.G., 2021, Standardization for the triple oxygen isotope system: Waters, silicates, carbonates, air, and sulfates: Reviews in Mineralogy and Geochemistry, v. 86, p. 179–196, https://doi.org/10.2138/rmg.2021.86.05.

Sharp, Z.D., Wostbrock, J.A.G., and Pack, A., 2018, Mass-dependent triple oxygen isotope variations in terrestrial materials: Geochemical Perspectives Letters, v. 7, p. 27–31, https://doi.org/10.7185/geochemlet.1815.

Shenton, B.J., Grossman, E.L., Passey, B.H., Henkes, G.A., Becker, T.P., Laya, J.C., Perez-Huerta, A., Becker, S.P., and Lawson, M., 2015, Clumped isotope thermometry in deeply buried sedimentary carbonates: The effects of bond reordering and recrystallization: Geological Society of America Bulletin, v. 127, p. 1036–1051, https://doi.org/10.1130/B31169.1.

Simon, R.E., Budd, D.A., and Snell, K.E., 2023, The history of calcite diagenesis and origin of exceptionally negative oxygen isotope values in chalks of the Niobrara Formation, Denver Basin, USA: The Depositional Record: The Journal of Biological, Physical and Geochemical Sedimentary Processes, v. 9, p. 607–634, https://doi.org/10.1002/dep2.218.

Slattery, J.S., Cobban, W.A., McKinney, K.C., Harries, P.J., and Sandness, A.L., 2015, Early Cretaceous to Paleocene paleogeography of the Western Interior Seaway: The interaction of eustasy and tectonism, *in* Bingle-Davis, M., ed., Field Guide, Cretaceous Conference: Evolution and Revolution: Casper, Wyoming, Wyoming Geological Association, p. 22–60.

Slattery, J.S., Harries, P.J., and Sandness, A.L., 2018, Do marine faunas track lithofacies?: Faunal dynamics in the Upper Cretaceous Pierre Shale, Western Interior, USA: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 496, p. 205–224, https://doi.org/10.1016/j.palaeo.2018.01.038.

Spötl, C., and Vennemann, T.W., 2003, Continuous-flow isotope ratio mass spectrometric analysis of carbonate minerals: Rapid Communications in Mass Spectrometry, v. 17, p. 1004–1006, https://doi.org/10.1002/rcm.1010.

Stinnesbeck, W., Ifrim, C., and Salazar, C., 2012, The Last Cretaceous ammonites in Latin America: Acta Palaeontologica Polonica, v. 57, p. 717–728, https://doi.org/10.4202/app.2011.0042.

Stolper, D.A., Eiler, J.M., and Higgins, J.A., 2018, Modeling the effects of diagenesis on carbonate clumped-isotope values in deep- and shallow-water settings: Geochimica et Cosmochimica Acta, v. 227, p. 264–291, https://doi.org/10.1016/j.gca.2018.01.037.

Taylor, H.P., 1978, Oxygen and hydrogen isotope studies of plutonic granitic rocks: Earth and Planetary Science Letters, v. 38, p. 177–210, https://doi.org/10.1016/0012-821X(78)90131-0.

Taylor, J.D., Glover, E.A., and Williams, S.T., 2014, Diversification of chemosymbiotic bivalves: Origins and relationships of deeper water Lucinidae: Biological Journal of the Linnean Society, v. 111, p. 401–420, https://doi.org/10.1111/bij.12208.

Tierney, J.E., et al., 2020, Past climates inform our future: Science, v. 370, https://doi.org/10.1126/science.aay3701.




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Tobin, T.S., and Ward, P.D., 2015, Carbon isotope ($\delta^{13}C$) differences between Late Cretaceous ammonites and benthic mollusks from Antarctica: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 428, p. 50–57, https://doi.org/10.1016/j.palaeo.2015.03.034.

Tobin, T.S., Wilson, G.P., Eiler, J.M., and Hartman, J.H., 2014, Environmental change across a terrestrial Cretaceous–Paleogene boundary section in eastern Montana, USA, constrained by carbonate clumped isotope paleothermometry: Geology, v. 42, p. 351–354, https://doi.org/10.1130/G35262.1.

Tourtelot, H.A., and Rye, R.O., 1969, Distribution of oxygen and carbon isotopes in fossils of Late Cretaceous age, western interior region of North America: Geological Society of America Bulletin, v. 80, p. 1903–1922, https://doi.org/10.1130/0016-7606(1969)80[1903:DOOACI]2.0.CO;2.

Tsujita, C.J., 1995, Origin of concretion-hosted shell clusters in the Late Cretaceous Bearpaw Formation, Southern Alberta, Canada: Palaios, v. 10, p. 408–423, https://doi.org/10.2307/3515044.

Tsujita, C.J., and Westermann, G.E.G., 1998, Ammonoid habitats and habits in the Western Interior Seaway: A case study from the Upper Cretaceous Bearpaw Formation of southern Alberta, Canada: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 144, p. 135–160, https://doi.org/10.1016/S0031-0182(98)00090-X.

Waage, K.M., 1964, Origin of repeated fossiliferous concretion layers in the Fox Hills Formation: Kansas Geological Survey Bulletin, v. 169, p. 541–563.

Waage, K.M., 1968, The Type Fox Hills Formation, Cretaceous (Maestrichtian), South Dakota, Pt. 1, stratigraphy and paleoenvironments: Peabody Museum of Natural History Bulletin, v. 27, p. 1–175.

Wang, S.C., and Dodson, P., 2006, Estimating the diversity of dinosaurs: Proceedings of the National Academy of Sciences of the United States of America, v. 103, p. 13,601–13,605, https://doi.org/10.1073/pnas.0606028103.

Witts, J.D., Landman, N.H., Hopkins, M.J., and Myers, C.E., 2020, Evolutionary stasis, ecophenotypy and environmental controls on ammonite morphology in the Late Cretaceous (Maastrichtian) Western Interior Seaway, USA: Palaeontology, v. 63, p. 791–806, https://doi.org/10.1111/pala.12488.

Wostbrock, J.A.G., and Sharp, Z.D., 2021, Triple oxygen isotopes in silica–water and carbonate–water systems: Reviews in Mineralogy and Geochemistry, v. 86, p. 367–400, https://doi.org/10.2138/rmg.2021.86.11.

Wostbrock, J.A.G., Sharp, Z.D., Sanchez-Yanez, C., Reich, M., van den Heuvel, D.B., and Benning, L.G., 2018, Calibration and application of silica-water triple oxygen isotope thermometry to geothermal systems in Iceland and Chile: Geochimica et Cosmochimica Acta, v. 234, p. 84–97, https://doi.org/10.1016/j.gca.2018.05.007.

Wostbrock, J.A.G., Brand, U., Coplen, T.B., Swart, P.K., Carlson, S.J., Brearley, A.J., and Sharp, Z.D., 2020a, Calibration of carbonate-water triple oxygen isotope fractionation: Seeing through diagenesis in ancient carbonates: Geochimica et Cosmochimica Acta, v. 288, p. 369–388, https://doi.org/10.1016/j.gca.2020.07.045.

Wostbrock, J.A.G., Cano, E.J., and Sharp, Z.D., 2020b, An internally consistent triple oxygen isotope calibration of standards for silicates, carbonates and air relative to VSMOW2 and SLAP2: Chemical Geology, v. 533, https://doi.org/10.1016/j.chemgeo.2019.119432.

Wright, E.K., 1987, Stratification and paleocirculation of the Late Cretaceous Western Interior Seaway of North America: Geological Society of America Bulletin, v. 99, p. 480–490, https://doi.org/10.1130/0016-7606(1987)99<480:SAPOTL>2.0.CO;2.

Zatoń, M., and Marynowski, L., 2006, Ammonite fauna from uppermost Bajocian (Middle Jurassic) calcitic concretions from the Polish Jura—Biogeographical and taphonomical implications: Geobios, v. 39, p. 426–442, https://doi.org/10.1016/j.geobios.2005.02.001.

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